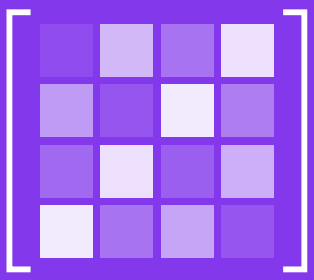
Linear Algebra, But Make It Make Sense

Lecture 2 – Matrices, linear maps and their properties



Sasha Sutton – personal study notes, Fall 2026

My notes + proofs + fully solved exercises, remixed into one doc.

How to read this doc (the box legend)

DEFINITION	The lore. Learn these word for word, examiners love them.
THEOREM	The canon. Things that are always true.
RECEIPTS	Proofs. Every claim needs receipts.
EXAMPLE	Worked examples with actual numbers.
ML SIDE QUEST	Why a data scientist should care.
EXAM CHEAT CODE	"If they ask you *this*, you do *that*". Step-by-step recipes.
COMMON L	Classic mistakes. Don't be that person.
TL;DR	The summary for when you're cramming at 2am.

Everything here is real maths, just explained like a friend would. The vibes are optional, the rigour is not.

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1 Previously on Linear Algebra (and today's notation)

The Lecture 1 recap you need

Today builds directly on Lecture 1. If any line below feels blurry, go back before reading on.

Concept	One-line reminder
Vector space, $\mathbb{R}^n$	A set where you can add and scale. $\mathbb{R}^n$ = lists of n real numbers.
Span	All linear combinations $c_1v_1 + \cdots + c_kv_k$. Always a subspace.
Independent / spanning	No vector is a combo of the others / the family generates everything.
Basis, dimension	Independent + spanning. Its size is the dimension. Gives *unique* coordinates.
Linear map $L : U \to V$	$L(au + bv) = aL(u) + bL(v)$. Determined by the images of a basis.
Kernel	$\ker L = \{x \mid L(x) = 0\}$: what gets crushed. Injective $\iff \ker L = \{0\}$.
Range, rank	$\operatorname{range} L = \{L(x)\}$: what's reachable. $\operatorname{rank} L = \dim \operatorname{range} L$. Surjective $\iff \operatorname{range} L = V$.
Rank–nullity	$\operatorname{rank} L + \dim \ker L = \dim U$ (dimension of the *domain*).

TL;DR // What changes today

Lecture 1 was abstract: linear maps as "functions that respect + and scaling". Today we make them **concrete**: once you pick bases, every linear map *is* a rectangle of numbers, a **matrix**. Everything from Lecture 1 (kernel, range, rank, injective, surjective) becomes something you can actually **compute**, by hand or on a computer.

Notation decoder

Symbol	Meaning
$\mathcal{M}_{m,n}(\mathbb{R})$ or $\mathbb{R}^{m\times n}$	matrices with m rows and n columns, real coefficients
a_{ij} or A_{ij}	the coefficient in **row** i, **column** j (row first, always)
$A^\top$	transpose: rows and columns swapped
I_n	identity matrix of size n
A^{-1}	inverse of A (when it exists)
$\operatorname{diag}(d_1, \ldots, d_n)$	diagonal matrix with $d_1, \ldots, d_n$ on the diagonal
$[x]_{\mathcal{B}}$	column of coordinates of the vector x in the basis $\mathcal{B}$
$[L]_{\mathcal{C}\leftarrow\mathcal{B}}$ or $[L]_{\mathcal{B},\mathcal{C}}$	matrix of L from basis $\mathcal{B}$ (input) to basis $\mathcal{C}$ (output)
$\ker A$, $\operatorname{range} A$	kernel and range of the map $x \mapsto Ax$. Also $\operatorname{Im} A$, $\operatorname{Col}(A)$ for the range
$[A \mid b]$	augmented matrix of the system $Ax = b$
$R_i \leftarrow R_i + \lambda R_j$	row operation: replace row i by row i plus λ times row j

All matrices in this lecture have **real** coefficients: $\mathcal{M}_{m,n}(\mathbb{R}) = \{A = (a_{ij})_{1\le i\le m,\, 1\le j\le n} \mid a_{ij} \in \mathbb{R}\}$.

2 Matrices and matrix operations

TL;DR // Chapter 2 in one breath

A matrix is a rectangle of numbers. You add them entry by entry, scale them entry by entry, and multiply them "row times column". Multiplication only works when the sizes fit, and the order matters: $AB \neq BA$ in general. The key mental image: Ax is a **linear combination of the columns of** A.

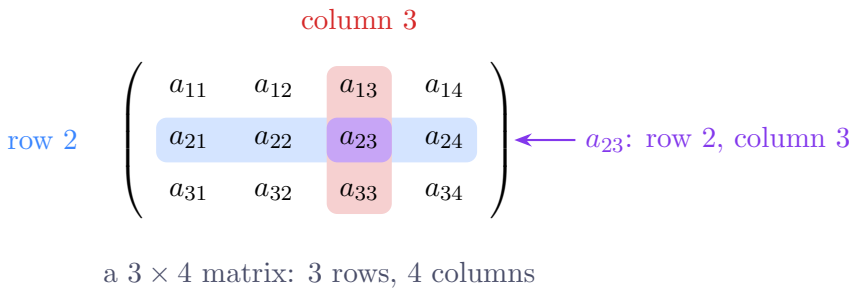
2.1 What is a matrix?

DEFINITION // Matrix

A **matrix** is a rectangular array of numbers called **coefficients** (or entries):

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}.$$

A has m **rows** and n **columns**: it is an $m \times n$ matrix. If $m = n$, A is a **square** matrix.



a 3×4 matrix: 3 rows, 4 columns

COMMON L (don't do this) // Rows first, columns second. Always.

"$m \times n$" = m rows, n columns. "a_{ij}" = row i, column j. Mnemonic: **R**oman **C**atholic, **RC** cola, **R**ows then **C**olumns. If you mix this up, every product you compute after that is cooked.

EXAMPLE // Reading a matrix

$A = \begin{pmatrix} 1 & 2 & 0 \\ 3 & -1 & 4 \end{pmatrix}$ is a 2×3 matrix (2 rows, 3 columns). $a_{12} = 2$, $a_{21} = 3$, $a_{23} = 4$. There is no a_{31}: the matrix has only 2 rows.
A vector $x \in \mathbb{R}^n$ is an $n \times 1$ matrix (a **column vector**). A $1 \times n$ matrix is a **row vector**.

2.2 Addition and scalar multiplication

DEFINITION // Sum and scalar multiple

For $A = (a_{ij})$ and $B = (b_{ij})$ **of the same size** $m \times n$, and $\lambda \in \mathbb{R}$:

$$A + B = (a_{ij} + b_{ij})_{1 \le i \le m,\, 1 \le j \le n}, \qquad \lambda A = (\lambda a_{ij})_{1 \le i \le m,\, 1 \le j \le n}.$$

EXAMPLE // Entry by entry

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} + \begin{pmatrix} 10 & 20 \\ 30 & 40 \end{pmatrix} = \begin{pmatrix} 11 & 22 \\ 33 & 44 \end{pmatrix}, \qquad 3 \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} = \begin{pmatrix} 3 & 6 \\ 9 & 12 \end{pmatrix}.$$

$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} + \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix}$ is **not defined** (sizes 2×2 and 2×3).

THEOREM / PROPOSITION // Matrices form a vector space

$(\mathcal{M}_{m,n}(\mathbb{R}), +, \cdot)$ is a real vector space of dimension mn.

RECEIPTS (proof) //

The operations are entry-wise, so every axiom reduces to the same axiom for real numbers (exactly like $\mathbb{R}^d$ in Lecture 1). The zero is the zero matrix, the opposite of A is $(-a_{ij})$. A basis is given by the mn matrices E_{ij} (a 1 in position (i, j), zeros elsewhere): any $A = \sum_{i=1}^{m} \sum_{j=1}^{n} a_{ij} E_{ij}$ (spanning), and if $\sum_{i,j} c_{ij} E_{ij} = 0$ then the entry (i, j) gives $c_{ij} = 0$ (independent). ■ *(and that's on periodt)*

A 28×28 image is a matrix, and $\mathcal{M}_{28,28}(\mathbb{R})$ has dimension 784: that's why Lecture 1 could flatten it into $\mathbb{R}^{784}$. Same vector space, different outfit.

2.3 Matrix times vector

DEFINITION // Matrix–vector product

For $A \in \mathbb{R}^{m \times n}$ and $x \in \mathbb{R}^n$, the product $Ax \in \mathbb{R}^m$ is

$$Ax = \begin{pmatrix} \sum_{j=1}^{n} a_{1j} x_j \\ \sum_{j=1}^{n} a_{2j} x_j \\ \vdots \\ \sum_{j=1}^{n} a_{mj} x_j \end{pmatrix}.$$

The number of **columns** of A must equal the number of **entries** of x.

There are two ways to see this product. Both give the same answer; the second one is the one that makes linear algebra click.

EXAMPLE // Two views of the same product

Let $A = \begin{pmatrix} 1 & 2 & 0 \\ 3 & -1 & 4 \end{pmatrix}$ and $x = \begin{pmatrix} 2 \\ 1 \\ -1 \end{pmatrix}$.

View 1: row by row (each output entry = one row of A "dotted" with x).

1. First entry: row 1 is $(1,2,0)$, so $1\cdot 2+2\cdot 1+0\cdot(-1)=4$.
2. Second entry: row 2 is $(3,-1,4)$, so $3\cdot 2+(-1)\cdot 1+4\cdot(-1)=1$.

So $Ax=\begin{pmatrix}4\\1\end{pmatrix}$.

View 2: column by column (a linear combination of the columns of A, weights = entries of x).

$$Ax = 2\begin{pmatrix}1\\3\end{pmatrix}+1\begin{pmatrix}2\\-1\end{pmatrix}+(-1)\begin{pmatrix}0\\4\end{pmatrix}=\begin{pmatrix}2+2-0\\6-1-4\end{pmatrix}=\begin{pmatrix}4\\1\end{pmatrix}. \checkmark$$

THEOREM / PROPOSITION // The column view

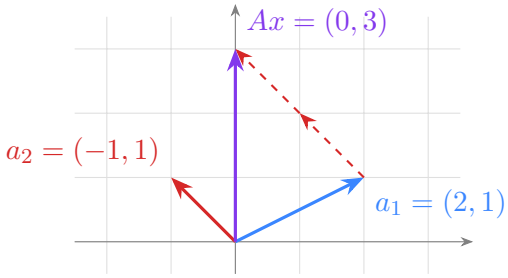
If $a_1,\dots,a_n\in\mathbb{R}^m$ are the columns of A, then

$$Ax = x_1a_1 + x_2a_2 + \cdots + x_na_n.$$

In particular $Ae_j = a_j$: multiplying by the j-th canonical vector **extracts column** j.

RECEIPTS (proof) //

The i-th entry of $x_1a_1+\cdots+x_na_n$ is $\sum_{j=1}^n x_ja_{ij}$, which is the i-th entry of Ax by definition. For $x=e_j$, all weights are 0 except $x_j=1$. ■ *(and that's on periodt)*



$$A=\begin{pmatrix}2&-1\\1&1\end{pmatrix},\quad x=\begin{pmatrix}1\\2\end{pmatrix}$$

$$Ax = 1\cdot a_1+2\cdot a_2 = (2,1)+(-2,2)=(0,3)$$

walk once along a_1, then twice along a_2.

TL;DR // Burn this into your brain

Ax is a linear combination of the columns of A. The entries of x are the recipe; the columns of A are the ingredients. So "all possible outputs Ax" = "the span of the columns of A". That one sentence explains range, rank and solvability of $Ax=b$ later.

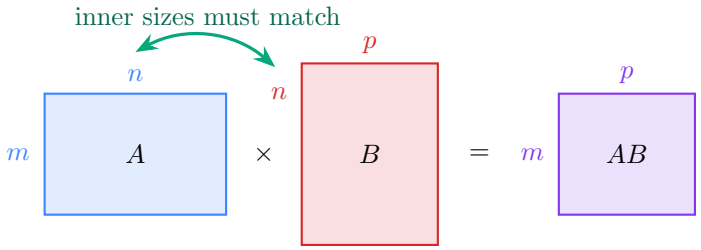
2.4 Matrix times matrix

DEFINITION // Matrix–matrix product

For $A\in\mathbb{R}^{m\times n}$ and $B\in\mathbb{R}^{n\times p}$, the product $AB\in\mathbb{R}^{m\times p}$ is $AB=(c_{ij})$ with

$$c_{ij}=\sum_{k=1}^{n}a_{ik}b_{kj}\qquad (1\le i\le m,\ 1\le j\le p).$$

In words: entry (i,j) of AB = (row i of A) times (column j of B).



$(m \times \underline{n}) \cdot (\underline{n} \times p) = (m \times p)$
the inner n's cancel,
the outer sizes survive.

EXAM CHEAT CODE // "Compute AB" (the algorithm)

1. **Check sizes.** Write them side by side: $(m \times n)(n' \times p)$. If $n \neq n'$: the product does not exist. Stop.
2. **Write the empty result** with m rows and p columns.
3. **Fill each entry** (i, j)**:** take row i of A and column j of B, multiply entry by entry, add up.
4. **Sanity check** one entry with the column view: column j of $AB = A \cdot$ (column j of B).

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} 5 & 6 \\ 7 & 8 \end{pmatrix} = \begin{pmatrix} 19 & 22 \\ 43 & 50 \end{pmatrix}$$

entry $(2,1)$: row 2 of A × column 1 of B
$3 \cdot 5 + 4 \cdot 7 = 15 + 28 = 43$

EXAMPLE // Full computation, all four entries

$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} 5 & 6 \\ 7 & 8 \end{pmatrix}$: sizes $(2 \times 2)(2 \times 2)$, fine, result is 2×2.

1. Entry $(1,1)$: row 1 × column 1 $= 1 \cdot 5 + 2 \cdot 7 = 19$.
2. Entry $(1,2)$: row 1 × column 2 $= 1 \cdot 6 + 2 \cdot 8 = 22$.
3. Entry $(2,1)$: row 2 × column 1 $= 3 \cdot 5 + 4 \cdot 7 = 43$.
4. Entry $(2,2)$: row 2 × column 2 $= 3 \cdot 6 + 4 \cdot 8 = 50$.

EXAMPLE // Non-square, and why order matters

$$A = \begin{pmatrix} 1 & 2 & 0 \\ 3 & -1 & 4 \end{pmatrix} (2 \times 3), \quad B = \begin{pmatrix} 1 & 0 \\ 2 & 1 \\ 0 & -1 \end{pmatrix} (3 \times 2).$$

AB: sizes $(2 \times \underline{3})(\underline{3} \times 2)$, result 2×2:

$$AB = \begin{pmatrix} 1 \cdot 1 + 2 \cdot 2 + 0 \cdot 0 & 1 \cdot 0 + 2 \cdot 1 + 0 \cdot (-1) \\ 3 \cdot 1 - 1 \cdot 2 + 4 \cdot 0 & 3 \cdot 0 - 1 \cdot 1 + 4 \cdot (-1) \end{pmatrix} = \begin{pmatrix} 5 & 2 \\ 1 & -5 \end{pmatrix}.$$

BA: sizes $(3 \times \underline{2})(\underline{2} \times 3)$, result 3×3:

$$BA = \begin{pmatrix} 1 & 2 & 0 \\ 5 & 3 & 4 \\ -3 & 1 & -4 \end{pmatrix}.$$

AB and BA don't even have the same size here. *"$AB = BA$" is not a thing you get to assume.*

THEOREM / PROPOSITION // Column view of the product

If $b_1, \dots, b_p$ are the columns of B, then $AB = \begin{bmatrix} Ab_1 \mid Ab_2 \mid \cdots \mid Ab_p \end{bmatrix}$: you apply A to each column of B. So matrix–matrix multiplication is just p matrix–vector multiplications stacked side by side.

RECEIPTS (proof) //

The i-th entry of Ab_j is $\sum_{k=1}^{n} a_{ik}b_{kj}$ (definition of matrix–vector product, where b_{kj} is the k-th entry of b_j). That is exactly c_{ij}. ■ *(and that's on periodt)*

THEOREM / PROPOSITION // Rules of matrix multiplication

For matrices of compatible sizes:

1. **Associativity:** $(AB)C = A(BC)$ for $A \in \mathbb{R}^{m\times n}$, $B \in \mathbb{R}^{n\times p}$, $C \in \mathbb{R}^{p\times q}$.
2. **Left distributivity:** $A(B+C) = AB + AC$ for $A \in \mathbb{R}^{m\times n}$, $B, C \in \mathbb{R}^{n\times p}$.
3. **Right distributivity:** $(A+B)C = AC + BC$ for $A, B \in \mathbb{R}^{m\times n}$, $C \in \mathbb{R}^{n\times p}$.
4. **Identity:** $I_m A = A$ and $AI_n = A$ for $A \in \mathbb{R}^{m\times n}$.
5. **Scalars move freely:** $\lambda(AB) = (\lambda A)B = A(\lambda B)$.

But multiplication is **not commutative**: $AB \neq BA$ in general.

RECEIPTS (proof) // index gymnastics

1. For $1 \le i \le m$, $1 \le j \le q$:

$$\big((AB)C\big)_{ij} = \sum_{l=1}^{p} (AB)_{il}c_{lj} = \sum_{l=1}^{p}\sum_{k=1}^{n} a_{ik}b_{kl}c_{lj} = \sum_{k=1}^{n} a_{ik}\sum_{l=1}^{p} b_{kl}c_{lj} = \sum_{k=1}^{n} a_{ik}(BC)_{kj} = \big(A(BC)\big)_{ij}$$

(We just swapped two finite sums.)
2. $\big(A(B+C)\big)_{ij} = \sum_{k=1}^{n} a_{ik}(b_{kj}+c_{kj}) = \sum_{k=1}^{n} a_{ik}b_{kj} + \sum_{k=1}^{n} a_{ik}c_{kj} = (AB)_{ij} + (AC)_{ij}$. *3.* is identical.
4. $(I_m A)_{ij} = \sum_{k=1}^{m} (I_m)_{ik}a_{kj} = a_{ij}$ because $(I_m)_{ik} = 1$ if $k = i$ and 0 otherwise. Same for AI_n.
5. Factor λ out of the sum. ■ *(and that's on periodt)*

EXAMPLE // Non-commutativity with a meaning

Let $M = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$ and $S = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ (the "swap" matrix).

$$MS = \begin{pmatrix} 2 & 1 \\ 4 & 3 \end{pmatrix} \text{ (columns of } M \text{ swapped)}, \qquad SM = \begin{pmatrix} 3 & 4 \\ 1 & 2 \end{pmatrix} \text{ (rows of } M \text{ swapped)}.$$

Multiplying on the right acts on **columns**; multiplying on the left acts on **rows**. Different operations, different results.

COMMON L (don't do this) // Matrices are not numbers: things that FAIL

1. $AB \neq BA$ in general. (Even when both products exist.)
2. $AB = 0$ does **not** imply $A = 0$ or $B = 0$: $\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}\begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$.
3. $AB = AC$ does **not** imply $B = C$ (no "simplifying by A" unless A is invertible).

4. $(A+B)^2 = A^2 + AB + BA + B^2$, **not** $A^2 + 2AB + B^2$ (unless $AB = BA$).
5. $(AB)^2 = ABAB$, **not** A^2B^2.

DEFINITION // Powers of a square matrix

For $A \in \mathbb{R}^{n\times n}$: $A^0 = I_n$, $A^1 = A$, $A^2 = AA$, $A^{k+1} = A^k A$. (Only square matrices have powers: AA needs the sizes to fit.)

3 Matrices as linear maps

TL;DR // Chapter 3 in one breath

A matrix is not just a table: it's a **linear map in disguise**. $x \mapsto Ax$ is always linear, and conversely every linear map between finite-dimensional spaces is "multiply by some matrix" once you choose bases. The recipe: **column j of the matrix = image of the j-th basis vector**. Composing maps = multiplying matrices.

3.1 Every matrix is a linear map

THEOREM / PROPOSITION // $x \mapsto Ax$ is linear

If $A \in \mathbb{R}^{m\times n}$, the map $L_A : \mathbb{R}^n \to \mathbb{R}^m$, $x \mapsto Ax$ is a linear transformation.

RECEIPTS (proof) //

Let $x, y \in \mathbb{R}^n$ and $\lambda, \mu \in \mathbb{R}$. For each row i $(1 \le i \le m)$:

$$\big(A(\lambda x + \mu y)\big)_i = \sum_{j=1}^{n} a_{ij}(\lambda x_j + \mu y_j) = \lambda \sum_{j=1}^{n} a_{ij}x_j + \mu \sum_{j=1}^{n} a_{ij}y_j = \lambda(Ax)_i + \mu(Ay)_i.$$

So $A(\lambda x + \mu y) = \lambda Ax + \mu Ay$. ■ *(and that's on periodt)*

COMMON L (don't do this) // Watch the dimensions

$A \in \mathbb{R}^{m\times n}$ goes **from** $\mathbb{R}^n$ **to** $\mathbb{R}^m$: the number of **columns** is the input dimension, the number of **rows** is the output dimension. A 2×3 matrix eats vectors of $\mathbb{R}^3$ and spits out vectors of $\mathbb{R}^2$. It reads "backwards". Everyone gets this wrong once.

3.2 Every linear map is a matrix (once you pick bases)

DEFINITION // Matrix of a linear map

Let V, W be finite-dimensional with bases $\mathcal{B} = (v_1, \dots, v_n)$ of V and $\mathcal{C} = (w_1, \dots, w_m)$ of W, and let $L : V \to W$ be linear. For each j, the vector $L(v_j) \in W$ has unique coordinates in $\mathcal{C}$:

$$L(v_j) = a_{1j}w_1 + \cdots + a_{mj}w_m = \sum_{i=1}^{m} a_{ij}w_i, \qquad j = 1, \dots, n.$$

The matrix $A = (a_{ij}) \in \mathbb{R}^{m\times n}$ is the **matrix of L in the bases $\mathcal{B}$ and $\mathcal{C}$**, written $[L]_{\mathcal{C}\leftarrow\mathcal{B}}$ or

$[L]_{\mathcal{B},\mathcal{C}}$.
The j-th column of A is the coordinate vector of $L(v_j)$ in the basis $\mathcal{C}$:

$$[L]_{\mathcal{C}\leftarrow\mathcal{B}} = \Big[\ [L(v_1)]_{\mathcal{C}} \ \Big|\ [L(v_2)]_{\mathcal{C}} \ \Big|\ \cdots \ \Big|\ [L(v_n)]_{\mathcal{C}}\ \Big].$$

THEOREM / PROPOSITION // The matrix does the job of the map

For every $x \in V$:

$$[L(x)]_{\mathcal{C}} = [L]_{\mathcal{C}\leftarrow\mathcal{B}}\,[x]_{\mathcal{B}}.$$

(coordinates of the output) = (matrix) × (coordinates of the input). Moreover this matrix is **unique**: for fixed bases, there's exactly one matrix that works.

RECEIPTS (proof) // existence + uniqueness

Formula. Let $x = \sum_{j=1}^{n} x_j v_j$, i.e. $[x]_{\mathcal{B}} = (x_1, \ldots, x_n)^{\top}$. By linearity,

$$L(x) = \sum_{j=1}^{n} x_j L(v_j) = \sum_{j=1}^{n} x_j \sum_{i=1}^{m} a_{ij} w_i = \sum_{i=1}^{m} \Big(\sum_{j=1}^{n} a_{ij} x_j \Big) w_i.$$

So the i-th coordinate of $L(x)$ in $\mathcal{C}$ is $\sum_{j=1}^{n} a_{ij} x_j$, the i-th entry of $A[x]_{\mathcal{B}}$.
Uniqueness. If A' also satisfies $[L(x)]_{\mathcal{C}} = A'[x]_{\mathcal{B}}$ for all x, take $x = v_j$: then $[v_j]_{\mathcal{B}} = e_j$, so $A'e_j = [L(v_j)]_{\mathcal{C}} = Ae_j$. But Ae_j is the j-th column of A. All columns agree, so $A = A'$. ■ *(and that's on periodt)*

TL;DR // The dictionary

With the **standard bases** of $\mathbb{R}^n$ and $\mathbb{R}^m$, coordinates are just the entries, so: every $A \in \mathbb{R}^{m\times n}$ gives the linear map $L(x) = Ax$, and every linear map $L : \mathbb{R}^n \to \mathbb{R}^m$ is $x \mapsto Ax$ for the unique matrix $A = [\,L(e_1) \mid \cdots \mid L(e_n)\,]$. Changing the basis in V or in W changes the matrix, *not* the map.

EXAM CHEAT CODE // "Find the matrix of T in the bases $\mathcal{B}$ and $\mathcal{C}$"

1. List the input basis vectors $v_1, \ldots, v_n$ (basis $\mathcal{B}$ of the domain).
2. Compute each image $T(v_1), \ldots, T(v_n)$.
3. Write each $T(v_j)$ as a combination of the output basis $\mathcal{C}$: $T(v_j) = a_{1j}w_1 + \cdots + a_{mj}w_m$. (If $\mathcal{C}$ is the standard basis, nothing to do: the coordinates are the entries.)
4. Put those coordinates in **column** j.
5. Check the size: (dim of codomain) rows × (dim of domain) columns.

Shortcut when T is given by a formula on $\mathbb{R}^n$ with standard bases: row i of the matrix = the coefficients of $x_1, \ldots, x_n$ in the i-th output coordinate.

EXAMPLE // From a formula to a matrix

$T : \mathbb{R}^3 \to \mathbb{R}^2$, $T(x, y, z) = (x + 2y,\ y - z)$, standard bases.

1. $T(e_1) = T(1, 0, 0) = (1, 0)$.
2. $T(e_2) = T(0, 1, 0) = (2, 1)$.
3. $T(e_3) = T(0, 0, 1) = (0, -1)$.

4. Columns: $A = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & -1 \end{pmatrix}$. Size 2×3: 2 rows (output in $\mathbb{R}^2$), 3 columns (input in $\mathbb{R}^3$). ✓
5. Check with the shortcut: first output $1x + 2y + 0z$ gives row $(1, 2, 0)$; second output $0x + 1y - 1z$ gives row $(0, 1, -1)$. ✓

EXAMPLE // Rotation of the plane

Rotation by angle θ (counter-clockwise) sends $e_1 = (1, 0)$ to $(\cos\theta, \sin\theta)$ and $e_2 = (0, 1)$ to $(-\sin\theta, \cos\theta)$. Columns = images of the basis:

$$R_\theta = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}. \qquad \text{For } \theta = 90^\circ: \; R = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \; R\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

EXAMPLE // Not just $\mathbb{R}^n$: the derivative is a matrix

Let $D : \mathbb{R}_2[X] \to \mathbb{R}_2[X]$, $D(P) = P'$ (derivative), with basis $\mathcal{B} = (1, X, X^2)$ on both sides.

1. $D(1) = 0 = 0 \cdot 1 + 0 \cdot X + 0 \cdot X^2$, coordinates $(0, 0, 0)$.
2. $D(X) = 1$, coordinates $(1, 0, 0)$.
3. $D(X^2) = 2X$, coordinates $(0, 2, 0)$.

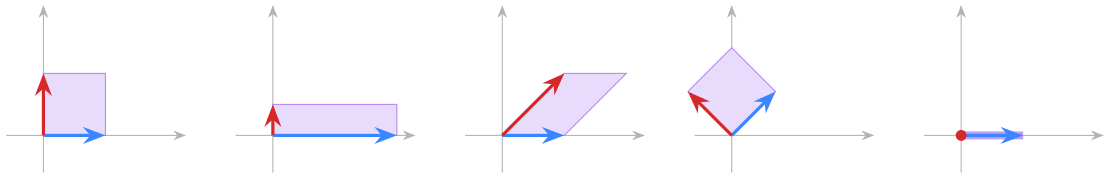
$$[D]_{\mathcal{B}} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{pmatrix}.$$

Test: $P = 3+5X+4X^2$ has coordinates $(3, 5, 4)$, and $\begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 3 \\ 5 \\ 4 \end{pmatrix} = \begin{pmatrix} 5 \\ 8 \\ 0 \end{pmatrix}$, i.e. $5+8X = P'$.

✓ *Calculus was linear algebra all along.*

3.3 Seeing a matrix: what it does to the plane

To visualise a 2×2 matrix, look at where it sends the unit square. The columns tell you everything: column 1 is where e_1 lands (blue), column 2 is where e_2 lands (red).



identity $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ | scaling $\begin{pmatrix} 2 & 0 \\ 0 & 0.5 \end{pmatrix}$ | shear $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ | rotation 45° $\frac{1}{\sqrt{2}}\begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$ | projection $\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$

The last one squashes the whole square onto a segment: e_2 is sent to 0. Information is destroyed, and that matrix will turn out to be *non-invertible*.

3.4 Composition of maps = product of matrices

THEOREM / PROPOSITION // Why matrix multiplication is defined that way

Let $B \in \mathbb{R}^{n\times p}$ and $A \in \mathbb{R}^{m\times n}$. Then for every $x \in \mathbb{R}^p$,

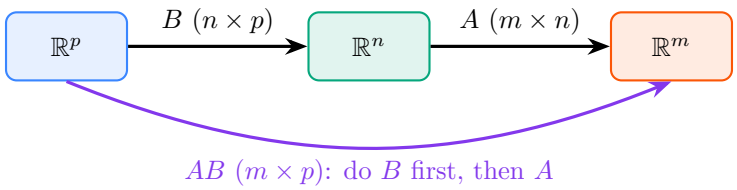
$$A(Bx) = (AB)x.$$

So the matrix of "first apply B, then apply A" is the product AB. More generally, for linear maps $L : U \to V$ and $M : V \to W$ with bases $\mathcal{B}, \mathcal{C}, \mathcal{D}$ of U, V, W:

$$[M \circ L]_{\mathcal{D}\leftarrow\mathcal{B}} = [M]_{\mathcal{D}\leftarrow\mathcal{C}}\,[L]_{\mathcal{C}\leftarrow\mathcal{B}}.$$

RECEIPTS (proof) // column view twice

Let $b_1, \ldots, b_p$ be the columns of B. Then $Bx = \sum_{j=1}^{p} x_j b_j$ (column view), and since $y \mapsto Ay$ is linear, $A(Bx) = \sum_{j=1}^{p} x_j Ab_j$. That is the linear combination of the vectors $Ab_1, \ldots, Ab_p$ with weights x_j, i.e. $[\,Ab_1 \mid \cdots \mid Ab_p\,]x = (AB)x$. The general statement follows by applying this to coordinate vectors: $[M(L(x))]_{\mathcal{D}} = [M]_{\mathcal{D}\leftarrow\mathcal{C}}[L(x)]_{\mathcal{C}} = [M]_{\mathcal{D}\leftarrow\mathcal{C}}[L]_{\mathcal{C}\leftarrow\mathcal{B}}[x]_{\mathcal{B}}$, and conclude by uniqueness of the matrix. ■ *(and that's on periodt)*



TL;DR // Three things this explains for free

1. **Why sizes must match:** the output of B (in $\mathbb{R}^n$) must be a valid input of A (which eats $\mathbb{R}^n$).
2. **Why it's associative:** composing functions is associative.
3. **Why it's not commutative:** "socks then shoes" $\neq$ "shoes then socks". AB means B first! Read products **right to left**.

EXAMPLE // Rotate then project vs project then rotate

$R = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ (rotation by 90°), $P = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$ (projection on the x-axis).

$$PR = \begin{pmatrix} 0 & -1 \\ 0 & 0 \end{pmatrix} \text{ (rotate, then project)}, \qquad RP = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} \text{ (project, then rotate)}.$$

On $e_1 = (1, 0)$: $PRe_1 = P(0, 1) = (0, 0)$ but $RPe_1 = R(1, 0) = (0, 1)$. Different maps, different matrices.

3.5 ML side quests

ML SIDE QUEST // Dense layers are linear maps (plus a shift)

A dense layer computes $x \in \mathbb{R}^d \mapsto Wx + b \in \mathbb{R}^k$ with $W \in \mathbb{R}^{k \times d}$, $b \in \mathbb{R}^k$.

- $x \mapsto Wx$ is **linear**; $x \mapsto Wx + b$ is **affine**.
- W stores the learned linear transformation; the bias b translates the output.
- Shape check: W is (output dim) $\times$ (input dim) $= k \times d$. Rows = neurons of the layer.

This is exactly `torch.nn.Linear(d, k)`.

ML SIDE QUEST // Why deep networks need non-linearities

Stack two dense layers *without* activation:

$$W_2(W_1x + b_1) + b_2 = \underbrace{(W_2W_1)}_{\text{one matrix}} x + \underbrace{(W_2b_1 + b_2)}_{\text{one bias}}.$$

It's still **one affine map**. By induction, 100 linear layers = 1 linear layer in a trench coat. The product W_2W_1 is matrix multiplication = composition. That's why we insert non-linear activations (ReLU, etc.) between layers: without them, depth buys nothing.

ML SIDE QUEST // Batches: matrix–matrix products

One input: $y = Wx$ with $W \in \mathbb{R}^{d \times d_{\text{in}}}$. For a **batch** of B inputs $x_1, \ldots, x_B$, stack them as columns:

$$X = [\,x_1 \mid x_2 \mid \cdots \mid x_B\,] \in \mathbb{R}^{d_{\text{in}} \times B}, \qquad Y = WX = [\,Wx_1 \mid Wx_2 \mid \cdots \mid Wx_B\,] \in \mathbb{R}^{d \times B}.$$

This is the column view of the product: one matrix–matrix multiplication processes the whole batch at once (and GPUs love that). *Note:* libraries like PyTorch usually store samples as **rows**, so they compute $Y = XW^\top$ instead; same maths, transposed.

4 Range, rank and kernel of a matrix

TL;DR // Chapter 4 in one breath

Take Lecture 1's range, rank and kernel and apply them to $x \mapsto Ax$. The **range** is the span of the columns. The **rank** is how many columns are genuinely independent (and, surprisingly, also how many rows are). The **kernel** is the set of x with $Ax = 0$: the input directions that A deletes. And **rank + dim kernel = number of columns**.

4.1 Column space, row space, rank

DEFINITION // Range / column space, row space

For $A \in \mathbb{R}^{m \times n}$:

- The **column space** (or **range**, or image) of A is the span of its columns, seen as vectors of $\mathbb{R}^m$:

$$\text{range}(A) = \text{Col}(A) = \{Ax \mid x \in \mathbb{R}^n\} = \text{span}(a_1, \ldots, a_n) \subseteq \mathbb{R}^m.$$

It is the range of the linear map $x \in \mathbb{R}^n \mapsto Ax \in \mathbb{R}^m$.

- The **row space** of A is the span of its rows, seen as vectors of $\mathbb{R}^n$. It is the range of the map $\mathbb{R}^m \to \mathbb{R}^n$ induced by $A^\top$.

The equality $\{Ax\} = \text{span}(\textit{columns})$ is the column view again: $Ax = x_1a_1 + \cdots + x_na_n$.

THEOREM / PROPOSITION // Row rank = column rank

The dimension of the column space of A equals the dimension of its row space. This common number is the **rank** of A:

$$\text{rank}(A) = \dim \text{range}(A) = \dim \text{range}(A^\top).$$

Consequences: $\text{rank}(A) = \text{rank}(A^\top)$ and $\text{rank}(A) \leq \min(m, n)$.

RECEIPTS (proof) // sketch, using Gaussian elimination

(The row operations and echelon forms used here are defined in Section 8.) Elementary row operations (swap two rows, add a multiple of a row to another, scale a row by $\lambda \neq 0$) have two properties:

1. They don't change the **row space**: new rows are combinations of old ones, and each operation can be undone.
2. They don't change the **solutions of** $Ax = 0$, i.e. they don't change $\ker A$. By rank–nullity, they don't change $\dim \text{range}(A) = n - \dim \ker(A)$ either.

So we may replace A by its reduced row-echelon form R, with r pivots. In R: the r non-zero rows are independent (each has a pivot 1 in a column where the others have 0), so the row space has dimension r. And $Rx = 0$ has $n - r$ free variables, so $\dim \ker = n - r$ and the column space has dimension $n - (n - r) = r$. Both equal the number of pivots. ■ *(and that's on periodt)*

EXAMPLE // Rank by inspection

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 1 & 0 & 1 \end{pmatrix}.$$

Columns: $a_3 = a_1 + a_2$ since $(1, 2, 1) + (2, 4, 0) = (3, 6, 1)$. So a_3 is redundant, and a_1, a_2 are not multiples of each other: column space has dimension 2.
Rows: row 2 = 2× row 1, and rows 1 and 3 are not multiples of each other: row space has dimension 2.
Both give $\text{rank}(A) = 2$. ✓ (As promised: same number, even though the redundancy looks completely different in rows and columns.)

4.2 Kernel

DEFINITION // Kernel / null space of a matrix

$\ker(A) = \{x \in \mathbb{R}^n \mid Ax = 0\} \subseteq \mathbb{R}^n$. Also written $\text{null}(A)$. It is the kernel of the map $x \mapsto Ax$, hence a subspace of $\mathbb{R}^n$ (Lecture 1).

THEOREM / PROPOSITION // Kernel, injectivity and independent columns

For $A \in \mathbb{R}^{m\times n}$, the following are equivalent:

1. $x \mapsto Ax$ is injective;
2. $\ker(A) = \{0\}$;
3. the columns of A are linearly independent.

RECEIPTS (proof) //

(1) $\iff$ (2) is Lecture 1 (a linear map is injective iff its kernel is $\{0\}$). (2) $\iff$ (3): by the column view, $Ax = 0$ reads $x_1a_1 + \cdots + x_na_n = 0$. So "the only solution of $Ax = 0$ is $x = 0$" is word for word "the only linear combination of the columns equal to 0 is the trivial one". ■ *(and that's on periodt)*

TL;DR // A kernel vector is a dependence relation

Every non-zero $x \in \ker(A)$ is a **recipe showing the columns are redundant**. In the example above, $a_1 + a_2 - a_3 = 0$, so $x = (1, 1, -1) \in \ker(A)$. Check: $A\begin{pmatrix}1\\1\\-1\end{pmatrix} = \begin{pmatrix}1+2-3\\2+4-6\\1+0-1\end{pmatrix} = \begin{pmatrix}0\\0\\0\end{pmatrix}$. ✓

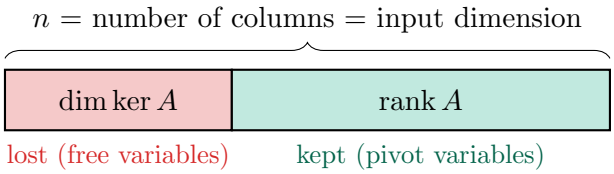
4.3 Rank–nullity for matrices

THEOREM / PROPOSITION // Rank–nullity theorem for matrices

For $A \in \mathbb{R}^{m\times n}$:

$$\boxed{n = \operatorname{rank}(A) + \dim\ker(A)}$$

where n is the number of **columns** of A (the dimension of the input space $\mathbb{R}^n$).



This is Lecture 1's theorem applied to $L : \mathbb{R}^n \to \mathbb{R}^m$, $L(x) = Ax$. In the example: $n = 3$, rank $= 2$, so $\dim\ker = 1$, and indeed $\ker(A) = \operatorname{span}\big((1, 1, -1)\big)$.

COMMON L (don't do this) // n is the number of COLUMNS

Rank–nullity uses the dimension of the **domain**, which for a matrix is the number of columns. Not the number of rows. For a 2×5 matrix of rank 2: $\dim\ker = 5 - 2 = 3$.

4.4 What the shape of a matrix tells you

Since $\operatorname{rank}(A) \le \min(m, n)$, the shape already decides a lot:

Shape of $A \in \mathbb{R}^{m\times n}$	Map $\mathbb{R}^n \to \mathbb{R}^m$	What's possible
Wide $(m < n)$	squashes into a smaller space	**Never injective**: $\dim\ker \geq n - m > 0$. Surjective iff rank $= m$.
Tall $(m > n)$	embeds into a bigger space	**Never surjective**: rank $\leq n < m$. Injective iff rank $= n$.
Square $(m = n)$	same dimension both sides	Injective $\iff$ surjective $\iff$ bijective $\iff$ rank $= n$.

THEOREM / PROPOSITION // Equivalent statements for a square matrix

If $A \in \mathbb{R}^{n\times n}$, the following are equivalent:

1. $x \mapsto Ax$ is bijective;
2. $\mathrm{Im}(A) = \mathbb{R}^n$ (surjective);
3. $\ker(A) = \{0\}$ (injective);
4. $\mathrm{rank}(A) = n$.

RECEIPTS (proof) // the rank–nullity ping-pong

(1) $\Rightarrow$ (2): bijective implies surjective. (2) $\Rightarrow$ (4): $\mathrm{rank}(A) = \dim \mathrm{Im}(A) = \dim \mathbb{R}^n = n$. (4) $\Rightarrow$ (3): by rank–nullity, $\dim\ker(A) = n - n = 0$, so $\ker(A) = \{0\}$. (3) $\Rightarrow$ (1): $\ker(A) = \{0\}$ gives injectivity, and $\mathrm{rank}(A) = n - 0 = n$, so $\mathrm{Im}(A)$ is an n-dimensional subspace of $\mathbb{R}^n$, hence all of $\mathbb{R}^n$: surjective too. ■ *(and that's on periodt)*

THEOREM / PROPOSITION // Rank of a product

For $A \in \mathbb{R}^{m\times n}$ and $B \in \mathbb{R}^{n\times p}$: $\quad \mathrm{rank}(AB) \leq \min\big(\mathrm{rank}(A), \mathrm{rank}(B)\big)$.

RECEIPTS (proof) //

Every $ABx = A(Bx)$ is of the form Ay, so $\mathrm{range}(AB) \subseteq \mathrm{range}(A)$ and $\mathrm{rank}(AB) \leq \mathrm{rank}(A)$. For the other bound, transpose: $\mathrm{rank}(AB) = \mathrm{rank}\big((AB)^\top\big) = \mathrm{rank}(B^\top A^\top) \leq \mathrm{rank}(B^\top) = \mathrm{rank}(B)$. ■ *(and that's on periodt)*

A chain is only as strong as its weakest link: once a matrix in the product has crushed some dimensions, no later matrix can bring them back.

4.5 ML side quests

ML SIDE QUEST // Range and kernel of a linear layer

For a layer $x \mapsto Wx$ with $W \in \mathbb{R}^{k\times d}$:

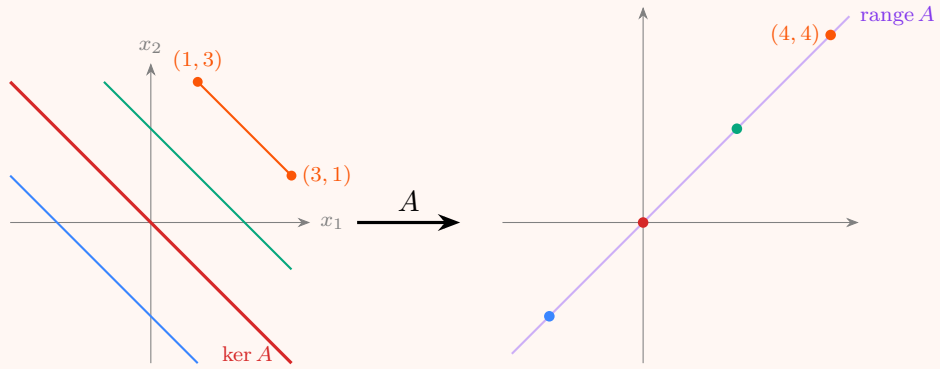
- $\mathrm{range}(W) = \{Wx : x \in \mathbb{R}^d\} \subseteq \mathbb{R}^k$: the set of outputs the layer **can** produce.
- $\ker(W) = \{x \in \mathbb{R}^d : Wx = 0\}$: the input directions the layer **ignores**.

ML SIDE QUEST // Kernel as lost information

$A = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$ gives $A \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} x_1 + x_2 \\ x_1 + x_2 \end{pmatrix}$.

- The matrix **keeps** the sum $x_1 + x_2$.
- It **loses** the difference $x_1 - x_2$.
- $\ker(A) = \operatorname{span}\left\{ \begin{pmatrix} 1 \\ -1 \end{pmatrix} \right\}$: solve $x_1 + x_2 = 0$, so $x = (t, -t) = t(1, -1)$.

Two inputs that differ by a kernel vector get the same output: $(3, 1)$ and $(1, 3)$ both go to $(4, 4)$, and $(3, 1) - (1, 3) = (2, -2) \in \ker(A)$.



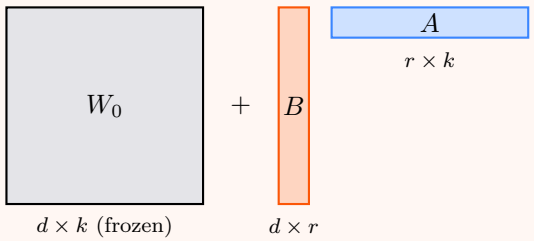
inputs: each line $x_1 + x_2 = c$ ... collapses to one point (c, c)

ML SIDE QUEST // LoRA: low-rank updates

Large networks contain huge weight matrices. Instead of fine-tuning every entry of $W_0 \in \mathbb{R}^{d \times k}$, **LoRA** learns

$$W_{\text{new}} = W_0 + \Delta W, \qquad \Delta W = BA, \qquad B \in \mathbb{R}^{d \times r},\ A \in \mathbb{R}^{r \times k},\ r \ll \min(d, k).$$

By the rank-of-a-product rule, $\operatorname{rank}(\Delta W) \leq \min(\operatorname{rank} B, \operatorname{rank} A) \leq r$. **LoRA restricts the trainable update to a low-dimensional subspace.**



full update: dk parameters
LoRA: $dr + rk = r(d + k)$ parameters

Parameter count for $W_0 \in \mathbb{R}^{4096 \times 4096}$ and $r = 8$:

1. Full update ΔW: $4096^2 = 16\,777\,216$ trainable parameters.
2. LoRA: $B \in \mathbb{R}^{4096 \times 8}$, $A \in \mathbb{R}^{8 \times 4096}$, so $4096 \cdot 8 + 8 \cdot 4096 = 65\,536$ parameters.
3. Ratio: $16\,777\,216 / 65\,536 = 256$. 256 **times fewer parameters.**

5 A few special matrices

TL;DR // Chapter 5 in one breath

Some matrices have a recognisable structure that makes them easy to work with: **diagonal** (just rescales each coordinate), **identity** (does nothing), **symmetric** (equal to its mirror image), plus sparse, triangular, band, orthogonal… The **transpose** flips rows and columns, and it reverses products: $(AB)^\top = B^\top A^\top$.

5.1 Diagonal matrices and the identity

DEFINITION // Diagonal matrix

A **diagonal matrix** is a square matrix whose coefficients outside the main diagonal are all zero:

$$D = \operatorname{diag}(d_1, \dots, d_n) = \begin{pmatrix} d_1 & 0 & \cdots & 0 \\ 0 & d_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & d_n \end{pmatrix}, \qquad d_1, \dots, d_n \in \mathbb{R}.$$

THEOREM / PROPOSITION // Why diagonal matrices are the easiest matrices

Let $D = \operatorname{diag}(d_1, \dots, d_n)$ and $D' = \operatorname{diag}(d'_1, \dots, d'_n)$.

1. $Dx = (d_1 x_1, \dots, d_n x_n)^\top$: D rescales each coordinate independently.
2. $DD' = \operatorname{diag}(d_1 d'_1, \dots, d_n d'_n) = D'D$: diagonal matrices multiply entry by entry and **commute**.
3. $D^k = \operatorname{diag}(d_1^k, \dots, d_n^k)$.
4. $D = D^\top$ (a diagonal matrix is symmetric).
5. D is invertible iff $d_i \neq 0$ for all i, and then $D^{-1} = \operatorname{diag}(1/d_1, \dots, 1/d_n)$.

RECEIPTS (proof) //

1. Row i of D has only one non-zero entry d_i in position i, so $(Dx)_i = d_i x_i$. *2.* $(DD')_{ij} = \sum_{k=1}^n D_{ik} D'_{kj}$; the term is non-zero only if $k = i$ and $k = j$, so we get $d_i d'_i$ when $i = j$ and 0 otherwise. Symmetric in D, D'. *3.* Induction using 2. *4.* Swapping i and j changes nothing off the diagonal (zeros) nor on it. *5.* If all $d_i \neq 0$, then by 2, $D \cdot \operatorname{diag}(1/d_i) = \operatorname{diag}(1) = I_n$ (both orders). If some $d_i = 0$, then $De_i = d_i e_i = 0$ with $e_i \neq 0$, so $\ker D \neq \{0\}$ and D is not injective, hence not invertible. ■ *(and that's on periodt)*

EXAMPLE // Diagonal in action

$D = \operatorname{diag}(2, 3)$: $D \begin{pmatrix} 5 \\ 1 \end{pmatrix} = \begin{pmatrix} 10 \\ 3 \end{pmatrix}$, $D^3 = \operatorname{diag}(8, 27)$, $D^{-1} = \operatorname{diag}(\frac{1}{2}, \frac{1}{3})$.

Multiplying another matrix by D: **on the left** it scales the **rows**, **on the right** it scales the **columns**:

$$\begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 2 \\ 3 & 3 \end{pmatrix}, \qquad \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix} = \begin{pmatrix} 2 & 3 \\ 2 & 3 \end{pmatrix}.$$

DEFINITION // Identity matrix

The **identity matrix** $I_n = \mathrm{diag}(1, \dots, 1)$ is the $n \times n$ diagonal matrix with ones on the diagonal.

THEOREM / PROPOSITION // Properties of the identity

For $A \in \mathbb{R}^{n \times n}$ and $x \in \mathbb{R}^n$: (1) $I_n x = x$; (2) $AI_n = I_n A = A$; (3) $I_n I_n = I_n$ and $(I_n)^{-1} = I_n$.

I_n is the "1" of matrix multiplication: the matrix of the map that does nothing.

ML SIDE QUEST // Diagonal matrices = feature rescaling

$D = \mathrm{diag}(d_1, \dots, d_n)$ acts coordinate by coordinate: it models **independent rescaling of features or channels**.

- **Feature normalisation:** standardising $x_j \mapsto (x_j - \mu_j)/\sigma_j$ is $x \mapsto D(x - \mu)$ with $D = \mathrm{diag}(1/\sigma_1, \dots, 1/\sigma_n)$.
- **Learned scales in normalisation layers:** batch normalisation multiplies each normalised channel by a learned γ_j, i.e. by $\mathrm{diag}(\gamma_1, \dots, \gamma_n)$.

5.2 Transpose

DEFINITION // Transpose

If $A \in \mathbb{R}^{m \times n}$, its **transpose** $A^\top \in \mathbb{R}^{n \times m}$ is defined by $(A^\top)_{ij} = A_{ji}$. The columns of A become the rows of $A^\top$, and the rows of A become the columns of $A^\top$.

EXAMPLE // Flip it

$A = \begin{pmatrix} 1 & 2 & 0 \\ 3 & -1 & 4 \end{pmatrix}$ $(2 \times 3) \Rightarrow A^\top = \begin{pmatrix} 1 & 3 \\ 2 & -1 \\ 0 & 4 \end{pmatrix}$ (3×2). The transpose of a column vector is a row vector, and the dot product from Lecture 1 is a matrix product: $x^\top y = \sum_{j=1}^n x_j y_j$ (a $1 \times n$ times an $n \times 1$ gives a 1×1).

THEOREM / PROPOSITION // Properties of the transpose

For matrices of compatible sizes and $\alpha \in \mathbb{R}$:

(i) $(A^\top)^\top = A$ (ii) $(A + B)^\top = A^\top + B^\top$ (iii) $(\alpha A)^\top = \alpha A^\top$ (iv) $(AB)^\top = B^\top A^\top$

RECEIPTS (proof) // the only non-obvious one is (iv)

(i)–(iii): swap indices twice / entry by entry. (iv): let $A \in \mathbb{R}^{m \times n}$, $B \in \mathbb{R}^{n \times p}$. For $1 \le i \le p$, $1 \le j \le m$:

$$\big((AB)^\top\big)_{ij} = (AB)_{ji} = \sum_{k=1}^{n} a_{jk} b_{ki} = \sum_{k=1}^{n} (B^\top)_{ik} (A^\top)_{kj} = (B^\top A^\top)_{ij}.$$

Size check: $B^\top$ is $p \times n$, $A^\top$ is $n \times m$, so $B^\top A^\top$ exists; $A^\top B^\top$ usually doesn't even make sense. ■ *(and that's on periodt)*

COMMON L (don't do this) // The order flips

$(AB)^\top = B^\top A^\top$, **not** $A^\top B^\top$. Same pattern as the inverse: $(AB)^{-1} = B^{-1}A^{-1}$. Socks and shoes again.

5.3 Symmetric matrices

DEFINITION // Symmetric, skew-symmetric

$A \in \mathbb{R}^{n\times n}$ is **symmetric** if $A = A^\top$, i.e. $A_{ij} = A_{ji}$ for all (i,j). It is **skew-symmetric** if $A = -A^\top$.

EXAMPLE // Mirror test

$\begin{pmatrix} 1 & 7 & 0 \\ 7 & 2 & -3 \\ 0 & -3 & 5 \end{pmatrix}$ is symmetric (mirror image across the diagonal). $\begin{pmatrix} 0 & 2 \\ -2 & 0 \end{pmatrix}$ is skew-symmetric (the diagonal of a skew-symmetric matrix is always 0 since $a_{ii} = -a_{ii}$). $\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$ is neither.

THEOREM / PROPOSITION // Free symmetric matrices

For **any** $A \in \mathbb{R}^{m\times n}$, the matrices $A^\top A \in \mathbb{R}^{n\times n}$ and $AA^\top \in \mathbb{R}^{m\times m}$ are symmetric. For any square A, $A = \underbrace{\tfrac{1}{2}(A + A^\top)}_{\text{symmetric}} + \underbrace{\tfrac{1}{2}(A - A^\top)}_{\text{skew-symmetric}}$.

RECEIPTS (proof) //

$(A^\top A)^\top = A^\top (A^\top)^\top = A^\top A$ by rule (iv) then (i). Same for $AA^\top$. For the decomposition: $\left(\tfrac{1}{2}(A + A^\top)\right)^\top = \tfrac{1}{2}(A^\top + A)$, and $\left(\tfrac{1}{2}(A - A^\top)\right)^\top = \tfrac{1}{2}(A^\top - A) = -\tfrac{1}{2}(A - A^\top)$; their sum is A. ■ *(and that's on periodt)*

EXAMPLE // A classic exercise: symmetric matrices that commute

Let $A, B \in \mathbb{R}^{n\times n}$ be symmetric and suppose AB is symmetric. (1) Show A and B commute. (2) Deduce that any two diagonal matrices commute.
(1) AB symmetric means $AB = (AB)^\top$. By rule (iv), $(AB)^\top = B^\top A^\top$. Since A, B are symmetric, $B^\top A^\top = BA$. Chain: $AB = (AB)^\top = B^\top A^\top = BA$. So A and B commute.
(2) Let D, D' be diagonal. Diagonal matrices are symmetric, and DD' is diagonal, hence symmetric. By (1), $DD' = D'D$.
(The converse of (1) also holds: if A, B symmetric and $AB = BA$ then $(AB)^\top = B^\top A^\top = BA = AB$.)

Teaser for Lecture 4 (spectral theorem): if $A \in \mathbb{R}^{n\times n}$ is symmetric, there is an orthonormal basis of $\mathbb{R}^n$ made of eigenvectors of A. In practice this gives the **eigendecomposition** of symmetric matrices. That's why symmetry is such a big deal.

ML SIDE QUEST // Symmetric matrices are everywhere in ML

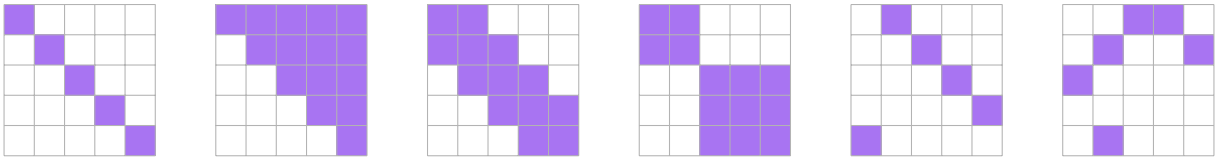
- **Covariance matrices.** For random variables $X_1, \ldots, X_n$ with finite variance, $K_{XX}(i,j) = \operatorname{cov}[X_i, X_j] = \mathbb{E}\big[(X_i - \mathbb{E}[X_i])(X_j - \mathbb{E}[X_j])\big]$. Symmetric because $\operatorname{cov}[X_i, X_j] = \operatorname{cov}[X_j, X_i]$. The empirical version on a centred data matrix X is $\frac{1}{m}X^\top X$: symmetric by

the theorem above.

- **Similarity matrices.** $S_{ij} = \text{sim}(x_i, x_j)$, symmetric whenever the similarity is (e.g. the Gram matrix $S_{ij} = x_i^\top x_j$).

Why it matters: real symmetric matrices admit an orthonormal eigenbasis (next lectures), which is the engine behind PCA.

5.4 Sparse, structured and other particular matrices



diagonal | upper triangular | band (tridiagonal) | block-diagonal | cyclic shift | sparse

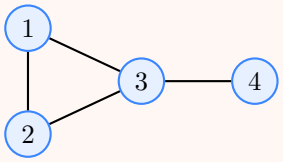
Coloured cells = entries allowed to be non-zero; white cells = zeros.

DEFINITION // A small zoo

- **Sparse matrix**: most coefficients are zero. You only store the non-zero ones.
- **Triangular matrix**: *upper* triangular if all entries below the diagonal are 0 ($a_{ij} = 0$ for $i > j$), *lower* triangular if all entries above are 0. Gaussian elimination produces these.
- **Band matrix**: non-zero entries only on a few diagonals around the main one.
- **Block-diagonal matrix**: square blocks on the diagonal, zeros elsewhere. It acts independently on groups of coordinates.
- **Matrix of ones** J: all entries equal to 1. $Jx = (\sum_i x_i)\,(1, \dots, 1)^\top$: useful for sums and means.
- **Shift matrix**: ones just above the diagonal; $S(x_1, x_2, x_3)^\top = (x_2, x_3, 0)^\top$. **Circulant matrix**: each row is the previous one shifted cyclically.
- **Orthogonal matrix**: $U^\top U = UU^\top = I$ (Section 6, and more in the next lectures).

ML SIDE QUEST // Sparse matrices in the wild

- **Bag-of-words / TF-IDF** matrices in NLP: one row per document, one column per word of the vocabulary; a document uses a tiny fraction of the vocabulary.
- **Attention masks.**
- **Recommender systems:** the user/item rating matrix, because a given user rates only a few items.
- **Adjacency matrices of graphs:** $A_{ij} = 1 \iff$ there is an edge between nodes i and j. Graph neural networks use this matrix to propagate information between neighbours.



$$A = \begin{pmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

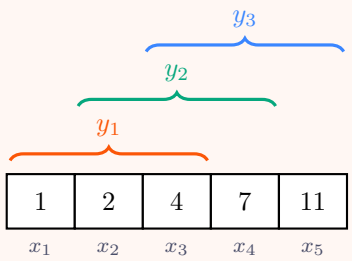
undirected graph
$\Rightarrow A$ symmetric.
Row 3 has three 1s:
node 3 has 3 neighbours.

ML SIDE QUEST // Toeplitz matrices and convolution

A one-dimensional convolution is a matrix multiplication. With kernel (a, b, c) sliding over $x \in \mathbb{R}^5$:

$$\begin{pmatrix} a & b & c & 0 & 0 \\ 0 & a & b & c & 0 \\ 0 & 0 & a & b & c \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{pmatrix} = \begin{pmatrix} ax_1 + bx_2 + cx_3 \\ ax_2 + bx_3 + cx_4 \\ ax_3 + bx_4 + cx_5 \end{pmatrix}.$$

The matrix has **repeated diagonals**: it is Toeplitz-like. Convolutional layers are not arbitrary dense matrices; they use strong structure: **local connectivity** (lots of zeros) and **shared coefficients** (the same a, b, c on every row). A dense 3×5 matrix has 15 parameters; this one has 3.



kernel $(a, b, c) = (1, -2, 1)$ slides over x:
$y_1 = 1 \cdot 1 - 2 \cdot 2 + 1 \cdot 4 = 1$
$y_2 = 1 \cdot 2 - 2 \cdot 4 + 1 \cdot 7 = 1$
$y_3 = 1 \cdot 4 - 2 \cdot 7 + 1 \cdot 11 = 1$

6 Invertibility and inverse matrices

TL;DR // Chapter 6 in one breath

A square matrix A is **invertible** if there's a matrix A^{-1} that undoes it: $A^{-1}A = AA^{-1} = I$. That happens exactly when $x \mapsto Ax$ loses no information: trivial kernel, full rank, independent columns, a unique solution to $Ax = b$... All of these are the *same* statement. Not every square matrix is invertible.

6.1 Definition

DEFINITION // Invertible matrix, inverse

Let $A \in \mathbb{R}^{n \times n}$. If there exists $B \in \mathbb{R}^{n \times n}$ such that

$$AB = BA = I_n,$$

then A is **invertible** (or **non-singular**) and B is the **inverse** of A, written A^{-1}. A non-invertible square matrix is called **singular**.
Equivalent definition (linear-map view): A is invertible iff the map $x \mapsto Ax$ is **bijective**; A^{-1} is then the matrix of the inverse map, and $A^{-1}A = AA^{-1} = I_n$.

THEOREM / PROPOSITION // The inverse is unique, and the two definitions agree

(1) If B and C are both inverses of A, then $B = C$. (2) $AB = BA = I_n$ for some $B \iff x \mapsto Ax$ is bijective.

RECEIPTS (proof) //

1. $B = BI_n = B(AC) = (BA)C = I_nC = C$.
2. ($\Rightarrow$) The map $y \mapsto By$ is an inverse function of $x \mapsto Ax$ (since $B(Ax) = x$ and $A(By) = y$), so $x \mapsto Ax$ is bijective. ($\Leftarrow$) If $L : x \mapsto Ax$ is bijective, its inverse L^{-1} is also linear: for $y, y' \in \mathbb{R}^n$, write $y = L(x)$, $y' = L(x')$; then $L^{-1}(\lambda y + \mu y') = L^{-1}(L(\lambda x + \mu x')) = \lambda x + \mu x' = \lambda L^{-1}(y) + \mu L^{-1}(y')$. So L^{-1} has a matrix B, and $L^{-1} \circ L = L \circ L^{-1} = \mathrm{Id}$ translates into $BA = AB = I_n$ (composition = product). ■ *(and that's on periodt)*

COMMON L (don't do this) // Inverse misconceptions

1. Only **square** matrices can be invertible (a bijection $\mathbb{R}^n \to \mathbb{R}^m$ linear forces $m = n$).
2. **Not every** square matrix is invertible: $\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$ is not.
3. A^{-1} is **not** "$1/a_{ij}$ entry by entry". That only works for diagonal matrices.
4. Never write $\frac{B}{A}$. Is it $A^{-1}B$ or BA^{-1}? They differ.

6.2 How to compute an inverse

EXAM CHEAT CODE // "Is A invertible? If so, compute A^{-1}"

Pick the fastest route:

1. **Diagonal** $\mathrm{diag}(d_1, \dots, d_n)$: invertible iff no d_i is zero; inverse $= \mathrm{diag}(1/d_1, \dots, 1/d_n)$.
2. 2×2: use the formula below.
3. **Orthogonal**: the inverse is the transpose.
4. **Anything else**: Gauss–Jordan on $[A \mid I_n]$ (Section 8). If the left block becomes I_n, the right block is A^{-1}; if a zero row appears on the left, A is not invertible.
5. **To show it's NOT invertible** quickly: find a non-zero x with $Ax = 0$, or spot two proportional rows/columns, a zero row/column, or a column that is a combination of the others.
6. **Always verify:** multiply $A \cdot A^{-1}$ and check you get I_n.

THEOREM / PROPOSITION // The 2×2 formula

$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ is invertible iff $ad - bc \neq 0$, and then

$$A^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$

Swap the diagonal, flip the signs of the other two, divide by $ad - bc$. (The number $ad - bc$ is the *determinant*: next lecture.)

RECEIPTS (proof) //

$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix} = \begin{pmatrix} ad - bc & -ab + ba \\ cd - dc & -cb + da \end{pmatrix} = (ad - bc)I_2$, and the same in the other order. If

$ad - bc \neq 0$, divide. If $ad - bc = 0$: either $A = 0$ (not invertible), or one of the vectors $(d, -c)$, $(-b, a)$ is a non-zero vector of $\ker A$ by the computation above, so A is not invertible. ■ *(and that's on periodt)*

EXAMPLE // 2×2 inverse step by step

$A = \begin{pmatrix} 2 & 1 \\ 5 & 3 \end{pmatrix}$.

1. $ad - bc = 2 \cdot 3 - 1 \cdot 5 = 1 \neq 0$: invertible.
2. Swap the diagonal: $\begin{pmatrix} 3 & \cdot \\ \cdot & 2 \end{pmatrix}$. Flip the signs of the others: $\begin{pmatrix} 3 & -1 \\ -5 & 2 \end{pmatrix}$.
3. Divide by 1: $A^{-1} = \begin{pmatrix} 3 & -1 \\ -5 & 2 \end{pmatrix}$.
4. Verify: $\begin{pmatrix} 2 & 1 \\ 5 & 3 \end{pmatrix}\begin{pmatrix} 3 & -1 \\ -5 & 2 \end{pmatrix} = \begin{pmatrix} 6-5 & -2+2 \\ 15-15 & -5+6 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$. ✓

6.3 Properties of invertible matrices

THEOREM / PROPOSITION // Algebra of inverses

If $A, B \in \mathbb{R}^{n \times n}$ are invertible and $\alpha \in \mathbb{R} \setminus \{0\}$:

1. $(A^{-1})^{-1} = A$
2. $(AB)^{-1} = B^{-1}A^{-1}$ (order reversed!)
3. $(A^\top)^{-1} = (A^{-1})^\top$
4. $(\alpha A)^{-1} = \frac{1}{\alpha}A^{-1}$
5. $I_n^{-1} = I_n$

RECEIPTS (proof) // just exhibit the inverse and check

To prove "$X^{-1} = Y$" it suffices to check $XY = YX = I_n$ (uniqueness of the inverse).
1. $A^{-1}A = AA^{-1} = I_n$ says precisely that A is the inverse of A^{-1}. *2.* $(AB)(B^{-1}A^{-1}) = A(BB^{-1})A^{-1} = AI_nA^{-1} = I_n$, and $(B^{-1}A^{-1})(AB) = B^{-1}(A^{-1}A)B = I_n$. *3.* $A^\top(A^{-1})^\top = (A^{-1}A)^\top = I_n^\top = I_n$, and $(A^{-1})^\top A^\top = (AA^{-1})^\top = I_n$. *4.* $(\alpha A)(\frac{1}{\alpha}A^{-1}) = \frac{\alpha}{\alpha}AA^{-1} = I_n$. *5.* $I_nI_n = I_n$. ■ *(and that's on periodt)*

COMMON L (don't do this) // Sums are a different story

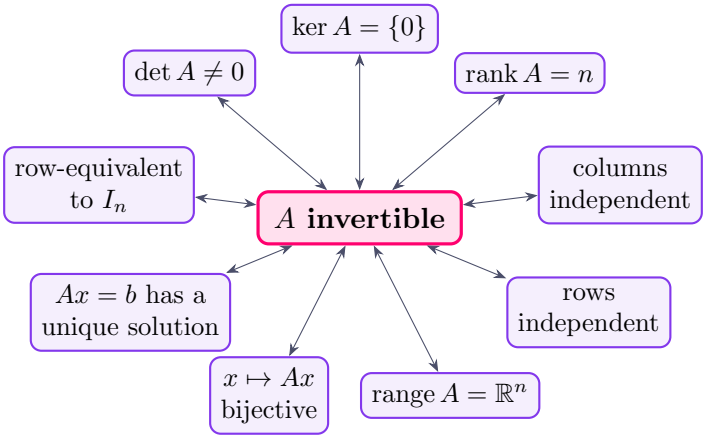
$(A + B)^{-1} \neq A^{-1} + B^{-1}$ in general, and $A + B$ may not even be invertible: I_n and $-I_n$ are invertible but $I_n + (-I_n) = 0$ is not.

6.4 The big theorem: ten ways to say "invertible"

THEOREM / PROPOSITION // Equivalent statements for an invertible matrix

Let $A \in \mathbb{R}^{n\times n}$. The following are equivalent:

1. **Invertibility:** A is invertible.
2. **Trivial kernel:** $\ker(A) = \{0\}$.
3. **Full rank:** $\operatorname{rank}(A) = n$.
4. The **columns** (resp. **rows**) of A are linearly independent.
5. **Span:** the columns of A span $\mathbb{R}^n$, i.e. $\operatorname{range}(A) = \mathbb{R}^n$.
6. **Linear map:** A is bijective as a linear transformation $\mathbb{R}^n \to \mathbb{R}^n$.
7. A is surjective as a linear transformation $\mathbb{R}^n \to \mathbb{R}^n$.
8. **Linear systems:** for any $b \in \mathbb{R}^n$, the equation $Ax = b$ has a unique solution.
9. **Row-equivalence:** A is row-equivalent to I_n (it can be turned into I_n by elementary row operations).
10. **Determinant:** $\det(A) \neq 0$ (next lecture).



RECEIPTS (proof) // how they chain together

(1) $\iff$ (6): the two definitions agree (previous theorem). (6) $\iff$ (7) $\iff$ (5) $\iff$ (3) $\iff$ (2): this is the "equivalent statements for a square matrix" theorem of Section 4 (surjective means range $A = \mathbb{R}^n$, i.e. rank $A = n$, i.e. $\dim \ker A = 0$ by rank–nullity, and for a square matrix injective $\iff$ surjective $\iff$ bijective).
(2) $\iff$ (4) for columns: $Ax = 0$ is a linear relation between the columns. For rows: apply the column statement to $A^\top$ and use $\operatorname{rank}(A^\top) = \operatorname{rank}(A)$.
(6) $\iff$ (8): "for every b there is exactly one x with $Ax = b$" is the definition of bijective.
(3) $\iff$ (9): the reduced row-echelon form of A has $\operatorname{rank}(A)$ pivots (Section 8); a square matrix has n pivots iff its reduced form is I_n.
(10): see the lecture on determinants. ■ *(and that's on periodt)*

TL;DR // How to use the big theorem

It's a menu. To prove A is invertible, pick the *easiest* item to check for your matrix. To use that A is invertible, pick the item you *need*. To show A is **not** invertible, break any single one (e.g. find $x \neq 0$ with $Ax = 0$).

THEOREM / PROPOSITION // A one-sided inverse is enough (for square matrices)

If $A, B \in \mathbb{R}^{n\times n}$ and $AB = I_n$, then A is invertible and $B = A^{-1}$ (so $BA = I_n$ automatically).

RECEIPTS (proof) //

For any $y \in \mathbb{R}^n$, $A(By) = y$: so $x \mapsto Ax$ is surjective, hence A is invertible (big theorem, item 7). Then $B = I_n B = (A^{-1}A)B = A^{-1}(AB) = A^{-1}$. ■ *(and that's on periodt)*

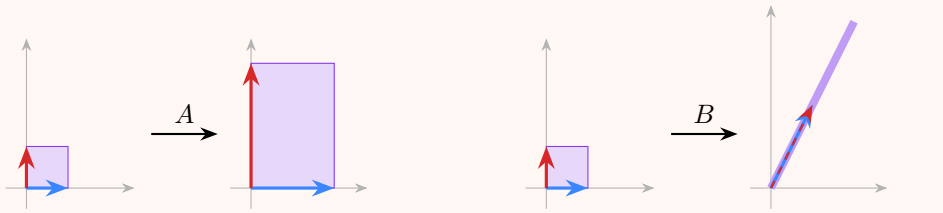
6.5 ML side quest: invertibility and information loss

ML SIDE QUEST // Invertible = no information lost

A square matrix is invertible exactly when $x \mapsto Ax$ loses no information: you can recover x from Ax.

Invertible: $A = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}$, $A^{-1} = \begin{pmatrix} 1/2 & 0 \\ 0 & 1/3 \end{pmatrix}$. The map stretches space; the inverse undoes the stretching.

Not invertible: $B = \begin{pmatrix} 1 & 1 \\ 2 & 2 \end{pmatrix}$ has rank 1 (second row = 2× first row; both columns equal): it collapses the whole plane onto the line spanned by $(1, 2)$. Many inputs give the same output, so there's no way back.



unit square stretched: still a 2D area collapsed onto a line

Some generative models (**normalizing flows**) are built entirely from invertible transformations, precisely so that nothing is lost and the map can be run backwards.

6.6 Orthogonal matrices (first look)

DEFINITION // Orthogonal matrix

A square matrix U is **orthogonal** if $U^\top U = UU^\top = I$. Equivalently, $U^{-1} = U^\top$.

THEOREM / PROPOSITION // What orthogonal means concretely

$U \in \mathbb{R}^{n\times n}$ is orthogonal iff its columns form an **orthonormal basis** of $\mathbb{R}^n$ (and the same holds for its rows). Moreover an orthogonal matrix preserves dot products and lengths: $(Ux)^\top(Uy) = x^\top y$.

RECEIPTS (proof) //

Let $u_1, \ldots, u_n$ be the columns of U. Entry (i, j) of $U^\top U$ is (row i of $U^\top$) × (column j of U) $= u_i^\top u_j$. So $U^\top U = I$ says exactly $u_i^\top u_j = 1$ if $i = j$ and 0 otherwise: orthonormal. An

orthonormal family is independent (Lecture 1 exercises), and n independent vectors in $\mathbb{R}^n$ form a basis. Conversely, orthonormal columns give $U^\top U = I$, and for a square matrix this one-sided identity already implies $UU^\top = I$ (one-sided inverse theorem). For rows, read $UU^\top = I$ the same way. Finally $(Ux)^\top(Uy) = x^\top U^\top U y = x^\top I y = x^\top y$. ■ *(and that's on periodt)*

EXAMPLE // Rotation matrices are orthogonal

$R_\theta = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$: $R_\theta^\top R_\theta = \begin{pmatrix} \cos^2\theta + \sin^2\theta & 0 \\ 0 & \sin^2\theta + \cos^2\theta \end{pmatrix} = I_2$. So $R_\theta^{-1} = R_\theta^\top = R_{-\theta}$: to undo a rotation by θ, rotate by $-\theta$. Makes sense.

For the determinant lecture: the determinant of a real orthogonal matrix is 1 or -1. *Link with Lecture 1:* the PCA encoder used a matrix $U \in \mathbb{R}^{d\times k}$ with orthonormal columns; then $U^\top U = I_k$ but $UU^\top \neq I_d$ when $k < d$ (it's the projection onto the subspace). "Orthogonal matrix" is reserved for the *square* case.

7 Change of basis and similarity

TL;DR // Chapter 7 in one breath

The same vector has different coordinates in different bases, and the same linear map has different matrices in different bases. The **change-of-basis matrix** P translates between the two coordinate systems, and the matrix of the map becomes $P^{-1}AP$. Two matrices related like this are **similar**: same map, different point of view. Choosing a clever basis can make an ugly matrix diagonal.

7.1 Changing coordinates

Let E be a vector space of dimension n with two bases: the "old" one $\mathcal{B} = (u_1, \dots, u_n)$ and the "new" one $\mathcal{B}' = (v_1, \dots, v_n)$.

DEFINITION // Change-of-basis matrix

The **change-of-basis matrix** P between $\mathcal{B}$ and $\mathcal{B}'$ is the matrix whose j-th column is $[v_j]_\mathcal{B}$, the coordinates of the **new** basis vector v_j written in the **old** basis $\mathcal{B}$:

$$P = P_{\mathcal{B}\leftarrow\mathcal{B}'} = \left[\, [v_1]_\mathcal{B} \;\middle|\; \cdots \;\middle|\; [v_n]_\mathcal{B} \,\right].$$

(Book notation: $P = \mathrm{P}_\mathcal{B}^{\mathcal{B}'}$. French: *matrice de passage de* $\mathcal{B}$ *à* $\mathcal{B}'$. It is also the matrix of the identity map, $[\mathrm{Id}_E]_{\mathcal{B}\leftarrow\mathcal{B}'}$.)

THEOREM / PROPOSITION // How coordinates change

For every $x \in E$:

$$[x]_\mathcal{B} = P\,[x]_{\mathcal{B}'} \qquad \text{and} \qquad [x]_{\mathcal{B}'} = P^{-1}[x]_\mathcal{B}.$$

P is invertible, and $P^{-1} = P_{\mathcal{B}'\leftarrow\mathcal{B}}$ is the matrix whose columns are the old basis vectors written in the new basis.

RECEIPTS (proof) //

Let $[x]_{\mathcal{B}'} = (c_1, \dots, c_n)^\top$, i.e. $x = \sum_{j=1}^n c_j v_j$. Taking coordinates in $\mathcal{B}$ is linear, so $[x]_{\mathcal{B}} = \sum_{j=1}^n c_j [v_j]_{\mathcal{B}}$: a linear combination of the columns of P with weights c_j, i.e. $P[x]_{\mathcal{B}'}$ (column view!).
Invertibility: swap the roles of $\mathcal{B}$ and $\mathcal{B}'$ to get $Q = P_{\mathcal{B}' \leftarrow \mathcal{B}}$ with $[x]_{\mathcal{B}'} = Q[x]_{\mathcal{B}}$. Then $[x]_{\mathcal{B}} = PQ[x]_{\mathcal{B}}$ for every x, and $[x]_{\mathcal{B}}$ runs over all of $\mathbb{R}^n$, so $PQ = I_n$; similarly $QP = I_n$. ■ *(and that's on periodt)*

COMMON L (don't do this) // The naming trap

The matrix built from the **new** basis vectors (written in the old basis) converts **new** coordinates into **old** coordinates: $[x]_{\text{old}} = P[x]_{\text{new}}$. To go old → new you need P^{-1}. It feels backwards; everyone trips on it. And the names clash: French textbooks call P "la matrice de passage de $\mathcal{B}$ à $\mathcal{B}'$", and other texts call $P^{-1} = \mathrm{P}_{\mathcal{B}'}^{\mathcal{B}}$ "the matrix to change from $\mathcal{B}$ to $\mathcal{B}'$" (because it turns $\mathcal{B}$-coordinates into $\mathcal{B}'$-coordinates). Same maths, opposite labels, so **always write down which formula your matrix satisfies**. The arrow notation $P_{\mathcal{B} \leftarrow \mathcal{B}'}$ is self-explaining: it eats $\mathcal{B}'$-coordinates and outputs $\mathcal{B}$-coordinates, and subscripts chain like dominoes: $P_{\mathcal{B} \leftarrow \mathcal{B}'} P_{\mathcal{B}' \leftarrow \mathcal{B}} = I_n$.

EXAM CHEAT CODE // "Find the coordinates of x in the new basis $\mathcal{B}'$"

In $\mathbb{R}^n$ with $\mathcal{B}$ the standard basis:

1. Build P: put the new basis vectors $v_1, \dots, v_n$ as columns.
2. Solve the linear system $Pc = x$ for c (or compute $c = P^{-1}x$).
3. $c = [x]_{\mathcal{B}'}$. Verify: $c_1 v_1 + \cdots + c_n v_n$ should give back x.

EXAMPLE // Coordinates in a new basis

In $\mathbb{R}^2$: $\mathcal{B}$ standard, $\mathcal{B}' = (v_1, v_2)$ with $v_1 = (1, 1)$, $v_2 = (1, -1)$. Take $x = (3, 1)$.

1. $P = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$.
2. Solve $c_1(1,1) + c_2(1,-1) = (3,1)$: $c_1 + c_2 = 3$ and $c_1 - c_2 = 1$. Adding: $2c_1 = 4$, so $c_1 = 2$, then $c_2 = 1$.
3. $[x]_{\mathcal{B}'} = (2, 1)$. Verify: $2(1,1) + 1(1,-1) = (3,1)$. ✓ And $P[x]_{\mathcal{B}'} = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 2 \\ 1 \end{pmatrix} = \begin{pmatrix} 3 \\ 1 \end{pmatrix} = [x]_{\mathcal{B}}$. ✓

With the 2×2 formula, $P^{-1} = \frac{1}{-2} \begin{pmatrix} -1 & -1 \\ -1 & 1 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$, and $P^{-1} \begin{pmatrix} 3 \\ 1 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 4 \\ 2 \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$. ✓

7.2 How the matrix of a map changes

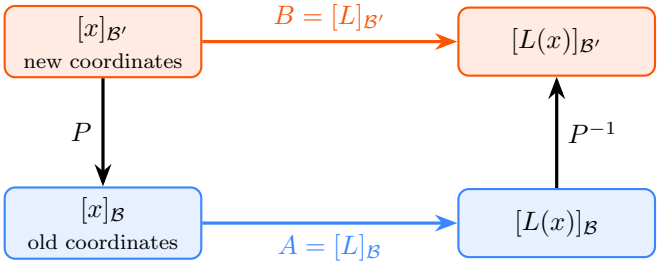
THEOREM / PROPOSITION // Theorem of basis change

Let $L : E \to E$ be a linear map, $A = [L]_{\mathcal{B}}$ its matrix in the basis $\mathcal{B}$ and $B = [L]_{\mathcal{B}'}$ its matrix in the basis $\mathcal{B}'$. With $P = P_{\mathcal{B} \leftarrow \mathcal{B}'}$ as above (so that $[x]_{\mathcal{B}} = P[x]_{\mathcal{B}'}$), P is invertible and

$$\boxed{B = P^{-1}AP} \qquad \text{i.e.} \qquad [L]_{\mathcal{B}'} = P_{\mathcal{B}' \leftarrow \mathcal{B}} \, [L]_{\mathcal{B}} \, P_{\mathcal{B} \leftarrow \mathcal{B}'}.$$

RECEIPTS (proof) // follow the coordinates

For any $x \in E$: $[L(x)]_{\mathcal{B}} = A[x]_{\mathcal{B}} = AP[x]_{\mathcal{B}'}$. Also $[L(x)]_{\mathcal{B}} = P[L(x)]_{\mathcal{B}'}$. Hence $P[L(x)]_{\mathcal{B}'} = AP[x]_{\mathcal{B}'}$, and multiplying by P^{-1}: $[L(x)]_{\mathcal{B}'} = P^{-1}AP\,[x]_{\mathcal{B}'}$ for all x. By uniqueness of the matrix of L in $\mathcal{B}'$, $B = P^{-1}AP$. ■ *(and that's on periodt)*



Top road: B.
Detour (down, across, up): $P^{-1}AP$. Same destination, so $B = P^{-1}AP$. Read right to left: P, then A, then P^{-1}.

EXAM CHEAT CODE // "Find the matrix of L in the new basis $\mathcal{B}'$"

Method 1 (definition): for each new basis vector v_j, compute $L(v_j)$, then write it in the basis $\mathcal{B}'$ (solve a small system). Those coordinates form column j.
Method 2 (formula):

1. Write $A = [L]_{\mathcal{B}}$ (matrix in the old basis).
2. Build P with the new basis vectors as columns (in old coordinates).
3. Compute P^{-1}.
4. Compute AP first, then $P^{-1}(AP)$.

Do both when you have time: they must agree.

EXAMPLE // A matrix that becomes diagonal in the right basis

$L : \mathbb{R}^2 \to \mathbb{R}^2$ with standard matrix $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$, and $\mathcal{B}' = (v_1, v_2)$, $v_1 = (1, 1)$, $v_2 = (1, -1)$.

Method 1.

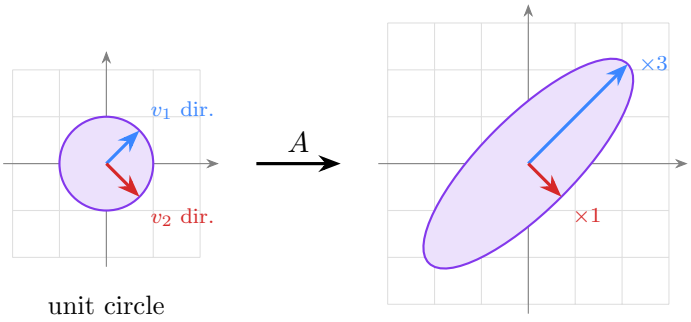
1. $L(v_1) = A\begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 3 \\ 3 \end{pmatrix} = 3v_1 + 0v_2$. Column 1: $(3, 0)$.
2. $L(v_2) = A\begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \end{pmatrix} = 0v_1 + 1v_2$. Column 2: $(0, 1)$.
3. $[L]_{\mathcal{B}'} = \begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix}$.

Method 2. $P = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$, $P^{-1} = \frac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$.

$$AP = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = \begin{pmatrix} 3 & 1 \\ 3 & -1 \end{pmatrix},$$

$$P^{-1}(AP) = \frac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}\begin{pmatrix} 3 & 1 \\ 3 & -1 \end{pmatrix} = \frac{1}{2}\begin{pmatrix} 6 & 0 \\ 0 & 2 \end{pmatrix} = \begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix}. \checkmark$$

In the basis $\mathcal{B}'$ the map is just "stretch by 3 along v_1, do nothing along v_2". Same map, way simpler description.



unit circle

stretched by 3 along $(1,1)$, unchanged along $(1,-1)$

7.3 Similar matrices

DEFINITION // Similar matrices (FR: matrices semblables)

Two **square** matrices $A, B \in \mathbb{R}^{n\times n}$ are **similar** if there exists an invertible $P \in \mathbb{R}^{n\times n}$ such that $B = P^{-1}AP$.
Interpretation: they represent the *same endomorphism* $L : V \to V$ in two different bases of V.

THEOREM / PROPOSITION // Things similar matrices share

If $B = P^{-1}AP$ then: (1) $\operatorname{rank}(B) = \operatorname{rank}(A)$; (2) B is invertible iff A is; (3) $B^k = P^{-1}A^kP$ for all $k \geq 0$.

RECEIPTS (proof) //

(1) A and B represent the same map L, and $\operatorname{rank} L = \dim \operatorname{range} L$ doesn't depend on any basis. (Or: $\operatorname{rank}(P^{-1}AP) \leq \operatorname{rank}(A)$ by the product rule, and $A = PBP^{-1}$ gives the other inequality.) (2) follows from (1). (3) $B^2 = P^{-1}A\underbrace{PP^{-1}}_{I}AP = P^{-1}A^2P$, and so on by induction: the inner PP^{-1} cancel like dominoes. ■ *(and that's on periodt)*

EXAMPLE // Why diagonal bases are OP: powers for free

From $D = P^{-1}AP$ we get $A = PDP^{-1}$ and $A^k = PD^kP^{-1}$. With $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$, $D = \operatorname{diag}(3,1)$:

$$A^5 = P\,\operatorname{diag}(3^5, 1^5)\,P^{-1} = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}\begin{pmatrix} 243 & 0 \\ 0 & 1 \end{pmatrix}\frac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = \frac{1}{2}\begin{pmatrix} 244 & 242 \\ 242 & 244 \end{pmatrix} = \begin{pmatrix} 122 & 121 \\ 121 & 122 \end{pmatrix}.$$

No need to multiply A by itself five times.

ML SIDE QUEST // Similarity as change of basis: PCA

Changing coordinates can make a problem easier without changing the underlying transformation. In PCA, a covariance matrix is expressed in a basis of eigenvectors, where it becomes **diagonal**:

$$\Sigma = U\Lambda U^\top.$$

The columns of U are the **principal directions**; the diagonal matrix Λ stores the variances along those directions. Here U is orthogonal, so $U^\top = U^{-1}$ and this reads $\Lambda = U^{-1}\Sigma U$: Σ and Λ are similar, with change-of-basis matrix $P = U$. The example above is a tiny version:

$A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$ is symmetric and becomes $\mathrm{diag}(3, 1)$ along the directions $(1, 1)$ and $(1, -1)$.

7.4 Similar vs equivalent matrices

What if $L : V \to W$ goes between two *different* spaces and we change the basis on both sides?

THEOREM / PROPOSITION // General change of bases

Let $L : V \to W$ be linear, $\mathcal{B}, \mathcal{B}'$ two bases of V and $\mathcal{C}, \mathcal{C}'$ two bases of W. Then

$$[L]_{\mathcal{C}' \leftarrow \mathcal{B}'} = P_{\mathcal{C}' \leftarrow \mathcal{C}} \, [L]_{\mathcal{C} \leftarrow \mathcal{B}} \, P_{\mathcal{B} \leftarrow \mathcal{B}'}.$$

RECEIPTS (proof) //

$[L(x)]_{\mathcal{C}'} = P_{\mathcal{C}' \leftarrow \mathcal{C}}[L(x)]_{\mathcal{C}} = P_{\mathcal{C}' \leftarrow \mathcal{C}}[L]_{\mathcal{C} \leftarrow \mathcal{B}}[x]_{\mathcal{B}} = P_{\mathcal{C}' \leftarrow \mathcal{C}}[L]_{\mathcal{C} \leftarrow \mathcal{B}} P_{\mathcal{B} \leftarrow \mathcal{B}'}[x]_{\mathcal{B}'}$, then uniqueness of the matrix. The subscripts chain like dominoes. ■ *(and that's on periodt)*

DEFINITION // Equivalent matrices (FR: matrices équivalentes)

$A, B \in \mathbb{R}^{m \times n}$ are **equivalent** if there exist invertible $Q \in \mathbb{R}^{m \times m}$ and $P \in \mathbb{R}^{n \times n}$ with $B = QAP$. **Interpretation:** they represent the same linear map after changing the basis in the domain *and* the basis in the codomain.

	Similar (semblables)	**Equivalent** (équivalentes)
Formula	$B = P^{-1}AP$	$B = QAP$
Sizes	square only ($n \times n$)	any $m \times n$
Meaning	same endomorphism $V \to V$, **one** basis changed (used on both sides)	same map $V \to W$, **two** independent basis changes
Relation	similar $\Rightarrow$ equivalent (take $Q = P^{-1}$)	equivalent $\not\Rightarrow$ similar

EXAMPLE // Equivalent but not similar

$I_2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ and $A = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}$ are equivalent ($I_2 = QAP$ with $Q = A^{-1}$, $P = I_2$). But the only matrix similar to I_2 is I_2 itself, because $P^{-1}I_2P = P^{-1}P = I_2$ for every P. So they are not similar. *(Fun fact, no proof needed here: two $m \times n$ matrices are equivalent iff they have the same rank.)*

8 Linear systems and Gaussian elimination

TL;DR // Chapter 8 in one breath

A linear system is $Ax = b$. It has **zero, one, or infinitely many** solutions, never exactly two. **Gaussian elimination** is the universal algorithm: use row operations to turn the system into a staircase (row-echelon form), then read off the answer. The same algorithm computes the rank, the kernel, a basis of the column space, and the inverse of a matrix. It is *the* tool of this lecture.

8.1 Linear systems and their matrix form

DEFINITION // Linear system

A system of m linear equations in n unknowns $x_1, \dots, x_n$:

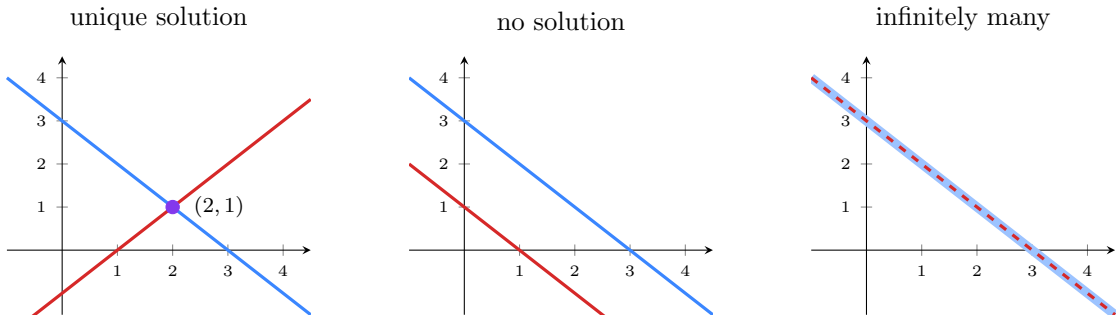
$$\begin{cases} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n = b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n = b_2 \\ \qquad\qquad\qquad\quad \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n = b_m \end{cases} \iff Ax = b$$

with $A = (a_{ij}) \in \mathbb{R}^{m\times n}$, $x = (x_1, \dots, x_n)^\top$, $b = (b_1, \dots, b_m)^\top$. The $a_{ij}, b_i \in \mathbb{R}$ are the **coefficients**. Every n-tuple $(x_1, \dots, x_n) \in \mathbb{R}^n$ satisfying *all* equations is a **solution**. The system is **homogeneous** if $b = 0$.

Column view: "solve $Ax = b$" means "find weights x_j such that b is that combination of the columns of A". So $Ax = b$ has a solution iff $b \in \text{range}(A)$.

8.2 Geometric picture in 2D

In dimension 2, each equation is a line in the plane; a solution is a point on *all* the lines.



EXAMPLE // The three cases, with numbers

1. **Unique solution:** $\begin{cases} x + y = 3 \\ x - y = 1 \end{cases}$ Adding: $2x = 4$, $x = 2$, $y = 1$. Lines intersect at one point.
2. **No solution:** $\begin{cases} x + y = 3 \\ x + y = 1 \end{cases}$ Subtracting: $0 = 2$. Impossible. Parallel disjoint lines.
3. **Redundancy, infinitely many solutions:** $\begin{cases} x + y = 3 \\ 2x + 2y = 6 \end{cases}$ The second equation is $2\times$ the first: same line. Solutions: $(x, y) = (3 - t, t)$, $t \in \mathbb{R}$.

8.3 The shape of the solution set

THEOREM / PROPOSITION // General solution = particular solution + kernel

Suppose x_p is one solution of $Ax = b$ (a **particular solution**). Then the set of all solutions is

$$\{x \in \mathbb{R}^n \mid Ax = b\} = \{x_p + h \mid h \in \ker(A)\} = x_p + \ker(A).$$

Consequently a linear system has **no** solution, **exactly one** (when $\ker A = \{0\}$), or **infinitely many**.

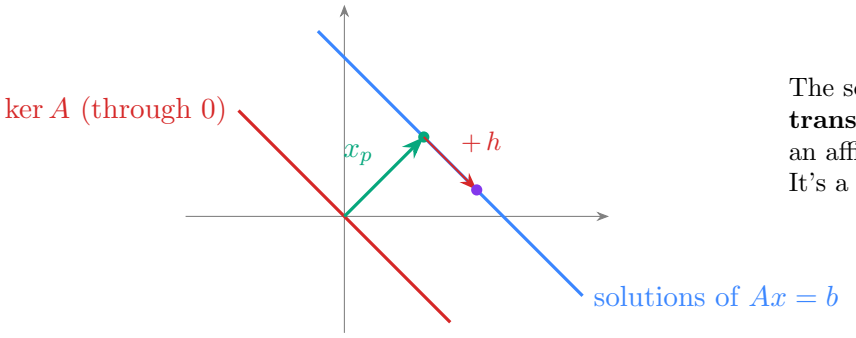
RECEIPTS (proof) //

($\supseteq$) If $h \in \ker A$: $A(x_p + h) = Ax_p + Ah = b + 0 = b$. ($\subseteq$) If $Ax = b$: $A(x - x_p) = b - b = 0$, so $h = x - x_p \in \ker A$ and $x = x_p + h$. For the count: if a solution exists, the solutions are in bijection with $\ker A$, a subspace, which is either $\{0\}$ or infinite. ■ *(and that's on periodt)*

EXAM CHEAT CODE // Method for the general solution of $Ax = b$

1. Find **one** particular solution x_p of $Ax = b$.
2. Find **all** solutions of the homogeneous system $Ax = 0$ (that's $\ker A$).
3. Add them: general solution $= x_p + \ker A$.

Note: a particular solution is generally not unique. The solution *set* is unique, but it can be parametrised in different ways (your friend's x_p may differ from yours and you can both be right).



The solution set is the kernel, **translated** by x_p: an affine subspace (Lecture 1). It's a subspace only if $b = 0$.

8.4 Elementary row operations

DEFINITION // Elementary transformations, augmented matrix

The **augmented matrix** of $Ax = b$ is $[A \mid b]$: the matrix A with the column b glued on the right. Each row is one equation.
The **elementary row operations** keep the solution set the same while making the system simpler:

1. **Swap** two rows (exchange two equations): $R_i \leftrightarrow R_j$.
2. **Add a multiple** of one row to another: $R_i \leftarrow R_i + \lambda R_j$ $(j \neq i)$.
3. **Scale** a row by a non-zero scalar: $R_i \leftarrow \lambda R_i$, $\lambda \in \mathbb{R}^\star$.

Two matrices are **row-equivalent** if one can be obtained from the other by a sequence of elementary row operations.

Why the solution set doesn't change: each operation is reversible (swap back; subtract what you added; divide by λ). Whatever satisfied the old equations satisfies the new ones, and vice versa.

COMMON L (don't do this) // Illegal moves

- Multiplying a row by 0 (you just deleted an equation).
- $R_i \leftarrow \lambda R_j$ *replacing* row i with a multiple of another row (row i's information is gone).
- Doing two operations "at once" that use each other's old values, e.g. $R_1 \leftarrow R_1 - R_2$ and $R_2 \leftarrow R_2 - R_1$ simultaneously. Do them one at a time.
- Applying the operation to A but forgetting the right-hand side b.

8.5 Row-echelon form

DEFINITION // Row-echelon form (REF)

A matrix is in **row-echelon form** if:

1. all rows containing only zeros are at the bottom;
2. looking at non-zero rows only, the **pivot** (first non-zero coefficient from the left, a.k.a. leading coefficient) of each row is strictly to the right of the pivot of the row above;
3. all entries below a pivot are zero.

DEFINITION // Reduced row-echelon form (RREF)

A matrix is in **reduced row-echelon form** (row canonical form) if:

1. it is in row-echelon form;
2. every pivot is 1;
3. the pivot is the only non-zero entry in its column.

Uniqueness: every $m \times n$ matrix is row-equivalent to a *unique* reduced row-echelon form.

$$\begin{pmatrix} \blacksquare & * & * & * & * & * \\ 0 & 0 & \blacksquare & * & * & * \\ 0 & 0 & 0 & \blacksquare & * & * \\ 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

REF: a staircase.
$\blacksquare$ = pivot ($\neq 0$), $*$ = anything

$$\begin{pmatrix} 1 & * & 0 & 0 & * & * \\ 0 & 0 & 1 & 0 & * & * \\ 0 & 0 & 0 & 1 & * & * \\ 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

RREF: pivots are 1 and alone in their (green) pivot columns

EXAMPLE // Echelon or not?

- $\begin{pmatrix} 2 & 1 & 3 \\ 0 & 0 & 5 \\ 0 & 0 & 0 \end{pmatrix}$: REF ✓ (pivots 2 and 5 form a staircase). Not reduced: the pivots are not 1, and there is a 3 above the pivot 5.
- $\begin{pmatrix} 1 & 2 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$: RREF ✓.

- $\begin{pmatrix} 1 & 2 & 3 \\ 0 & 0 & 0 \\ 0 & 1 & 4 \end{pmatrix}$: not REF (the zero row is not at the bottom).
- $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$: not REF (the second pivot is to the *left* of the first).

8.6 Gaussian elimination: the algorithm

EXAM CHEAT CODE // "Solve $Ax = b$" (Gaussian elimination)

1. **Write the augmented matrix** $[A \mid b]$.
2. **Forward elimination** (go left to right, top to bottom):
   (a) In the current column, find a non-zero entry at or below the current row. If there's none, move one column right. Otherwise swap it up: that's your **pivot**.
   (b) Kill every entry **below** the pivot: $R_i \leftarrow R_i - \dfrac{\text{entry}}{\text{pivot}} R_{\text{pivot row}}$.
   (c) Move one row down and one column right; repeat.

   You now have a row-echelon form.
3. **Check consistency:** a row $(0 \ \cdots \ 0 \mid c)$ with $c \neq 0$ means "$0 = c$": **no solution**. Stop.
4. **Identify variables:** columns with a pivot give **pivot variables**; the others are **free variables**.
5. **Back-substitution:** solve from the last non-zero row upwards, expressing pivot variables in terms of the free ones. (Or keep reducing to RREF and read the answer directly.)
6. **Write the solution** as $x_p + (\text{free variable}) \cdot (\text{vector}) + \dots$ and **check** it in the original equations.

Tip: swap rows to get a pivot equal to ± 1 when you can. It avoids fractions and saves your sanity.

EXAMPLE // Worked example 1: a unique solution

$$\begin{cases} x + y + z = 6 \\ 2x - y + z = 3 \\ x + 2y - z = 2 \end{cases} \quad \Longrightarrow \quad \left(\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 2 & -1 & 1 & 3 \\ 1 & 2 & -1 & 2 \end{array}\right)$$

Step 1: clear column 1 below the pivot 1 (row 1). $R_2 \leftarrow R_2 - 2R_1$ and $R_3 \leftarrow R_3 - R_1$:

$$\left(\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & -3 & -1 & -9 \\ 0 & 1 & -2 & -4 \end{array}\right)$$

Step 2: choose the pivot in column 2. Swap $R_2 \leftrightarrow R_3$ to get a pivot 1 (no fractions):

$$\left(\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & -2 & -4 \\ 0 & -3 & -1 & -9 \end{array}\right)$$

Step 3: clear column 2 below the pivot. $R_3 \leftarrow R_3 + 3R_2$:

$$\left(\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & -2 & -4 \\ 0 & 0 & -7 & -21 \end{array}\right) \qquad \text{(row-echelon form, 3 pivots)}$$

Step 4: back-substitution, from the bottom up.

1. Row 3: $-7z = -21$, so $z = 3$.
2. Row 2: $y - 2z = -4$, so $y = -4 + 6 = 2$.
3. Row 1: $x + y + z = 6$, so $x = 6 - 2 - 3 = 1$.

Step 5: check in the original system: $1 + 2 + 3 = 6$ ✓, $2 - 2 + 3 = 3$ ✓, $1 + 4 - 3 = 2$ ✓. Unique solution $(x, y, z) = (1, 2, 3)$. Three pivots for three unknowns: no free variable, $\ker A = \{0\}$.

EXAMPLE // Worked example 2: infinitely many solutions

$$\begin{cases} x_1 + 2x_2 + x_3 + x_4 = 4 \\ 2x_1 + 4x_2 + 3x_3 + x_4 = 9 \\ x_1 + 2x_2 + 2x_3 = 5 \end{cases} \quad \Longrightarrow \quad \left(\begin{array}{cccc|c} 1 & 2 & 1 & 1 & 4 \\ 2 & 4 & 3 & 1 & 9 \\ 1 & 2 & 2 & 0 & 5 \end{array}\right)$$

Step 1: clear column 1. $R_2 \leftarrow R_2 - 2R_1$, $R_3 \leftarrow R_3 - R_1$:

$$\left(\begin{array}{cccc|c} 1 & 2 & 1 & 1 & 4 \\ 0 & 0 & 1 & -1 & 1 \\ 0 & 0 & 1 & -1 & 1 \end{array}\right)$$

Step 2: column 2 has only zeros below row 1: no pivot there, move right. Pivot in column 3 (row 2). $R_3 \leftarrow R_3 - R_2$:

$$\left(\begin{array}{cccc|c} 1 & 2 & 1 & 1 & 4 \\ 0 & 0 & 1 & -1 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right) \qquad \text{(REF: pivots in columns 1 and 3)}$$

Step 3: consistency. The last row says $0 = 0$: fine. **Pivot variables:** x_1, x_3. **Free variables:** x_2, x_4.

Step 4: reduce further (clear above the pivot of column 3). $R_1 \leftarrow R_1 - R_2$:

$$\left(\begin{array}{cccc|c} 1 & 2 & 0 & 2 & 3 \\ 0 & 0 & 1 & -1 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right) \qquad \text{(RREF)}$$

Step 5: read off. Row 1: $x_1 = 3 - 2x_2 - 2x_4$. Row 2: $x_3 = 1 + x_4$. Set $x_2 = s$, $x_4 = t$:

$$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \underbrace{\begin{pmatrix} 3 \\ 0 \\ 1 \\ 0 \end{pmatrix}}_{x_p} + s \underbrace{\begin{pmatrix} -2 \\ 1 \\ 0 \\ 0 \end{pmatrix}}_{h_1} + t \underbrace{\begin{pmatrix} -2 \\ 0 \\ 1 \\ 1 \end{pmatrix}}_{h_2}, \qquad s, t \in \mathbb{R}.$$

Step 6: check. x_p: $3 + 0 + 1 + 0 = 4$ ✓, $6 + 0 + 3 + 0 = 9$ ✓, $3 + 0 + 2 = 5$ ✓. h_1 in the homogeneous system: $-2 + 2 = 0$ ✓, $-4 + 4 = 0$ ✓, $-2 + 2 = 0$ ✓. h_2: $-2 + 1 + 1 = 0$ ✓, $-4 + 3 + 1 = 0$ ✓, $-2 + 2 = 0$ ✓.

This is exactly "particular solution + kernel": $\ker A = \operatorname{span}(h_1, h_2)$, $\operatorname{rank} A = 2$, and $2 + 2 = 4$ columns. ✓

EXAMPLE // Worked example 3: no solution

Same system, but the last right-hand side is 6 instead of 5. The same operations give

$$\left(\begin{array}{cccc|c} 1 & 2 & 1 & 1 & 4 \\ 2 & 4 & 3 & 1 & 9 \\ 1 & 2 & 2 & 0 & 6 \end{array}\right) \xrightarrow[R_3 \leftarrow R_3 - R_1]{R_2 \leftarrow R_2 - 2R_1} \left(\begin{array}{cccc|c} 1 & 2 & 1 & 1 & 4 \\ 0 & 0 & 1 & -1 & 1 \\ 0 & 0 & 1 & -1 & 2 \end{array}\right)$$

$$\xrightarrow{R_3 \leftarrow R_3 - R_2} \left(\begin{array}{cccc|c} 1 & 2 & 1 & 1 & 4 \\ 0 & 0 & 1 & -1 & 1 \\ 0 & 0 & 0 & 0 & 1 \end{array}\right)$$

The last row reads $0x_1 + 0x_2 + 0x_3 + 0x_4 = 1$, i.e. $0 = 1$. **Inconsistent: no solution.** (A pivot appeared in the b column.)

8.7 Rank and the number of solutions

THEOREM / PROPOSITION // Pivots, rank and solutions

The number of pivots in a row-echelon form of A equals $\operatorname{rank}(A)$ (the maximum number of linearly independent rows, or columns, of A). For $A \in \mathbb{R}^{m \times n}$ and the system $Ax = b$:

Condition	Solutions	What it looks like
$\operatorname{rank}(A) < \operatorname{rank}([A \mid b])$	**none** (inconsistent)	a row $(0 \ \cdots \ 0 \mid c)$, $c \neq 0$
$\operatorname{rank}(A) = \operatorname{rank}([A \mid b]) = n$	**exactly one**	a pivot in every column of A
$\operatorname{rank}(A) = \operatorname{rank}([A \mid b]) < n$	**infinitely many**	$n - \operatorname{rank}(A)$ free variables

The number of free variables is the **nullity** $\dim \ker(A) = n - \operatorname{rank}(A)$ (rank–nullity again).

RECEIPTS (proof) // why

$Ax = b$ is solvable iff b is a combination of the columns of A, iff adding the column b to A doesn't increase the dimension of the column space, iff $\operatorname{rank}([A \mid b]) = \operatorname{rank}(A)$. When solvable, the solution set is $x_p + \ker A$ with $\dim \ker A = n - \operatorname{rank} A$: a single point iff $\operatorname{rank} A = n$. ■ *(and that's on periodt)*

In the three worked examples: (1) $\operatorname{rank} A = \operatorname{rank}[A \mid b] = 3 = n$: one solution. (2) $\operatorname{rank} A = \operatorname{rank}[A \mid b] = 2 < 4$: infinitely many, 2 free variables. (3) $\operatorname{rank} A = 2 < 3 = \operatorname{rank}[A \mid b]$: none.

8.8 Reading everything off the echelon form

THEOREM / PROPOSITION // What the reduced form gives you

Let R be the reduced row-echelon form of A.

1. $\operatorname{rank}(A)$ = number of pivots.
2. The **pivot columns of the original** A form a basis of the column space of A.
3. The **non-zero rows of** R form a basis of the row space of A.
4. Solving $Rx = 0$ with the free variables as parameters gives a basis of $\ker(A)$ (one vector per free variable).

COMMON L (don't do this) // Column space: go back to the ORIGINAL matrix

Row operations preserve the row space and the kernel, but they **change the column space**. So for a basis of $\mathrm{Col}(A)$: use the echelon form only to learn *which* columns are pivot columns, then take those columns **from** A, not from R.

EXAM CHEAT CODE // "Find the rank, a basis of the column space and a basis of the kernel of A"

1. Row-reduce A (no right-hand side needed, or think $b = 0$).
2. Count the pivots: that's $\mathrm{rank}(A)$.
3. Note the pivot columns; take the same columns of the **original** A: basis of $\mathrm{Col}(A)$.
4. Free variables = non-pivot columns. For each free variable in turn: set it to 1, the other free variables to 0, and solve for the pivot variables. You get one kernel vector per free variable: basis of $\ker(A)$.
5. Sanity check: $\mathrm{rank}(A) + \dim\ker(A) =$ number of columns, and $A \cdot$(each kernel vector) $= 0$.

EXAMPLE // All four at once

$$A = \begin{pmatrix} 1 & 2 & 0 & 1 & 3 \\ 2 & 4 & 1 & 3 & 8 \\ 1 & 2 & 1 & 2 & 5 \end{pmatrix}.$$

Step 1: row-reduce. $R_2 \leftarrow R_2 - 2R_1$, $R_3 \leftarrow R_3 - R_1$, then $R_3 \leftarrow R_3 - R_2$:

$$\begin{pmatrix} 1 & 2 & 0 & 1 & 3 \\ 0 & 0 & 1 & 1 & 2 \\ 0 & 0 & 1 & 1 & 2 \end{pmatrix} \longrightarrow R = \begin{pmatrix} 1 & 2 & 0 & 1 & 3 \\ 0 & 0 & 1 & 1 & 2 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} \quad \text{(already reduced)}.$$

Step 2: rank. Two pivots (columns 1 and 3): $\mathrm{rank}(A) = 2$.
Step 3: column space. Columns 1 and 3 of the *original* A: $\mathrm{Col}(A) = \mathrm{span}\big((1,2,1),(0,1,1)\big)$.
Step 4: row space. Non-zero rows of R: $\mathrm{span}\big((1,2,0,1,3),(0,0,1,1,2)\big)$.
Step 5: kernel. Free variables x_2, x_4, x_5. From R: $x_1 = -2x_2 - x_4 - 3x_5$ and $x_3 = -x_4 - 2x_5$.

- $x_2 = 1,\ x_4 = x_5 = 0$: $(-2,1,0,0,0)$.
- $x_4 = 1,\ x_2 = x_5 = 0$: $(-1,0,-1,1,0)$.
- $x_5 = 1,\ x_2 = x_4 = 0$: $(-3,0,-2,0,1)$.

$\ker(A) = \mathrm{span}\big((-2,1,0,0,0),(-1,0,-1,1,0),(-3,0,-2,0,1)\big)$, dimension 3.
Step 6: check. $2+3 = 5$ columns ✓. And the kernel vectors are dependence relations between columns: the first says $a_2 = 2a_1$; the second says $a_4 = a_1 + a_3$, indeed $(1,2,1)+(0,1,1) = (1,3,2)$ ✓; the third says $a_5 = 3a_1 + 2a_3 = (3,8,5)$ ✓.

8.9 Computing an inverse by Gaussian elimination

EXAM CHEAT CODE // Gauss–Jordan: $[A \mid I_n] \to [I_n \mid A^{-1}]$

1. Write the augmented matrix $[A \mid I_n]$ (the matrix A with the identity glued on the right).
2. Apply row operations **to the whole row** (both blocks) to bring the left block to *reduced* row-echelon form: first clear below the pivots (top to bottom), make each pivot 1, then clear above the pivots (bottom to top).

3. If the left block becomes I_n: the right block is A^{-1}.
4. If a zero row appears in the left block: stop, A is **not invertible**.
5. Verify $AA^{-1} = I_n$.

RECEIPTS (proof) // why it works

Finding A^{-1} means finding X with $AX = I_n$. Column by column (column view of the product), this is n linear systems $Ax_j = e_j$ with the *same* matrix A. Gauss–Jordan solves them all at once: the right block carries the n right-hand sides $e_1, \dots, e_n$. When the left block has become I_n, the systems read $I_n x_j =$ (column j of the right block), so the right block is $X = A^{-1}$ (a right inverse of a square matrix is the inverse). If the left block cannot be reduced to I_n, then A has fewer than n pivots, rank $A < n$, and A is not invertible. So **computing an inverse is equivalent to solving linear systems**. ■ *(and that's on periodt)*

EXAMPLE // Gauss–Jordan on a 2×2 (compare with the formula)

$A = \begin{pmatrix} 2 & 1 \\ 5 & 3 \end{pmatrix}$:

$$\left(\begin{array}{cc|cc} 2 & 1 & 1 & 0 \\ 5 & 3 & 0 & 1 \end{array}\right) \xrightarrow{R_1 \leftarrow \frac{1}{2}R_1} \left(\begin{array}{cc|cc} 1 & \frac{1}{2} & \frac{1}{2} & 0 \\ 5 & 3 & 0 & 1 \end{array}\right) \xrightarrow{R_2 \leftarrow R_2 - 5R_1} \left(\begin{array}{cc|cc} 1 & \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & \frac{1}{2} & -\frac{5}{2} & 1 \end{array}\right)$$

$$\xrightarrow{R_2 \leftarrow 2R_2} \left(\begin{array}{cc|cc} 1 & \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 1 & -5 & 2 \end{array}\right) \xrightarrow{R_1 \leftarrow R_1 - \frac{1}{2}R_2} \left(\begin{array}{cc|cc} 1 & 0 & 3 & -1 \\ 0 & 1 & -5 & 2 \end{array}\right)$$

So $A^{-1} = \begin{pmatrix} 3 & -1 \\ -5 & 2 \end{pmatrix}$, the same as with the 2×2 formula. ✓

EXAMPLE // Gauss–Jordan on a 3×3

$M = \begin{pmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 1 & 3 & 3 \end{pmatrix}$.

$$\left(\begin{array}{ccc|ccc} 1 & 2 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 1 & 3 & 3 & 0 & 0 & 1 \end{array}\right) \xrightarrow{R_3 \leftarrow R_3 - R_1} \left(\begin{array}{ccc|ccc} 1 & 2 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & 2 & -1 & 0 & 1 \end{array}\right)$$

$$\xrightarrow{R_3 \leftarrow R_3 - R_2} \left(\begin{array}{ccc|ccc} 1 & 2 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & -1 & -1 & 1 \end{array}\right)$$

Three pivots: M is invertible. Now clear *above* the pivots, starting from the last column:

$$\xrightarrow[R_1 \leftarrow R_1 - R_3]{R_2 \leftarrow R_2 - R_3} \left(\begin{array}{ccc|ccc} 1 & 2 & 0 & 2 & 1 & -1 \\ 0 & 1 & 0 & 1 & 2 & -1 \\ 0 & 0 & 1 & -1 & -1 & 1 \end{array}\right) \xrightarrow{R_1 \leftarrow R_1 - 2R_2} \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 0 & -3 & 1 \\ 0 & 1 & 0 & 1 & 2 & -1 \\ 0 & 0 & 1 & -1 & -1 & 1 \end{array}\right)$$

$$M^{-1} = \begin{pmatrix} 0 & -3 & 1 \\ 1 & 2 & -1 \\ -1 & -1 & 1 \end{pmatrix}.$$

Check (first row of MM^{-1}): $(0 + 2 - 1,\ -3 + 4 - 1,\ 1 - 2 + 1) = (1, 0, 0)$. ✓ The other two rows work the same way.

EXAMPLE // When it fails

$A = \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}$: $\left(\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 2 & 4 & 0 & 1 \end{array}\right) \xrightarrow{R_2 \leftarrow R_2 - 2R_1} \left(\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 0 & 0 & -2 & 1 \end{array}\right)$. A zero row on the left: only one pivot, $\operatorname{rank} A = 1 < 2$, **not invertible**. (Makes sense: row $2 = 2\times$ row 1.)

8.10 ML side quests

ML SIDE QUEST // Solving a linear system = fitting a model exactly

Linear model $f(t) = at + b$. We want the line to pass through $(1, 2)$ and $(3, 6)$:

$$\begin{cases} a + b = 2 \\ 3a + b = 6 \end{cases} \iff \begin{pmatrix} 1 & 1 \\ 3 & 1 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 2 \\ 6 \end{pmatrix}.$$

$R_2 \leftarrow R_2 - 3R_1$ gives $-2b = 0$, so $b = 0$, then $a = 2$. The model is $f(t) = 2t$. *This is supervised learning in its simplest exact form:* unknowns = parameters, one equation per data point.

ML SIDE QUEST // Least squares: when there are too many equations

Usually we have more data points than parameters: $X\beta \approx y$ with $X \in \mathbb{R}^{n \times d}$, $n > d$ (a *tall* matrix, an **overdetermined** system). There is typically **no exact solution** to $X\beta = y$, because $y \notin \operatorname{range}(X)$. Instead, linear regression solves

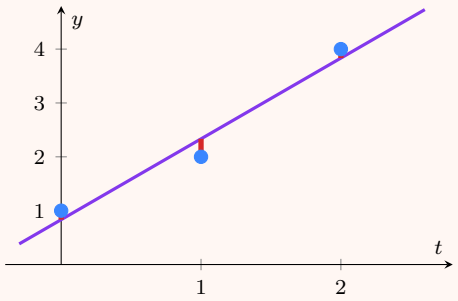
$$\hat{\beta} = \arg\min_{\beta \in \mathbb{R}^d} \|X\beta - y\|_2^2, \qquad \text{and when } X^\top X \text{ is invertible:} \quad \hat{\beta} = (X^\top X)^{-1} X^\top y.$$

Equivalently, $\hat{\beta}$ solves the square system $X^\top X\beta = X^\top y$ (the *normal equations*).

Tiny example. Fit $f(t) = at + b$ to the points $(0, 1), (1, 2), (2, 4)$ (not aligned, so no exact fit).

1. One row per point, columns $= (t, 1)$: $X = \begin{pmatrix} 0 & 1 \\ 1 & 1 \\ 2 & 1 \end{pmatrix}$, $y = \begin{pmatrix} 1 \\ 2 \\ 4 \end{pmatrix}$, $\beta = \begin{pmatrix} a \\ b \end{pmatrix}$.
2. $X^\top X = \begin{pmatrix} 0^2 + 1^2 + 2^2 & 0 + 1 + 2 \\ 0 + 1 + 2 & 1 + 1 + 1 \end{pmatrix} = \begin{pmatrix} 5 & 3 \\ 3 & 3 \end{pmatrix}$, $X^\top y = \begin{pmatrix} 0 \cdot 1 + 1 \cdot 2 + 2 \cdot 4 \\ 1 + 2 + 4 \end{pmatrix} = \begin{pmatrix} 10 \\ 7 \end{pmatrix}$.
3. Solve $5a + 3b = 10$, $3a + 3b = 7$. Subtract: $2a = 3$, so $a = \frac{3}{2}$; then $b = \frac{7 - 4.5}{3} = \frac{5}{6}$.

Best-fit line: $f(t) = 1.5\,t + 0.83$.



Red segments: the residuals (gaps between each point and the line). Least squares makes the sum of their squares as small as possible.

Link with Lecture 1. $X^\top X$ is invertible iff the columns of X are linearly independent. Indeed $\ker(X^\top X) = \ker(X)$: if $X^\top Xv = 0$ then $v^\top X^\top Xv = \|Xv\|^2 = 0$, so $Xv = 0$. So redundant features (dependent columns) make $X^\top X$ singular: that's the "design matrix becomes singular" problem from Lecture 1, now with receipts.

9 The exercises, fully solved

TL;DR // How to use this section

Try each question first, then check. Each solution is written the way you'd write it in an exam: say what you're doing, show the steps, box the answer, verify. The extra steps and remarks below are here to show *how* to get there.

9.1 Part 1(a): Short exercises

Exercise 1

Find the matrix of $T : \mathbb{R}^3 \to \mathbb{R}^3$ *defined by* $T(e_1) = (1,0,0)^\top$, $T(e_2) = (0,1,0)^\top$, $T(e_3) = (0,0,1)^\top$, *where* (e_1, e_2, e_3) *is the standard basis.*

SOLUTION // Ex. 1

The columns of the matrix of a linear map in the standard basis are the images of the standard basis vectors: $[T] = [\,T(e_1) \mid T(e_2) \mid T(e_3)\,]$. Here $T(e_1) = e_1$, $T(e_2) = e_2$, $T(e_3) = e_3$, so

$$[T] = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = I_3.$$

T is the identity transformation of $\mathbb{R}^3$. *Free points: columns = images of the basis.*

Exercise 2

$T : \mathbb{R}^2 \to \mathbb{R}^2$, $T(x_1, x_2) = (x_1, 2x_2)$. *(1) Matrix in the standard basis. (2) Matrix in the basis* $B' = (v_1, v_2)$, $v_1 = (1,1)^\top$, $v_2 = (1,-1)^\top$.

SOLUTION // Ex. 2.1 – standard basis

$T(e_1) = T(1,0) = (1,0)$ and $T(e_2) = T(0,1) = (0,2)$. Columns:

$$[T]_{\text{std}} = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}.$$

SOLUTION // Ex. 2.2 – basis B', method 1 (definition)

We compute $T(v_1)$ and $T(v_2)$, then express each in the basis (v_1, v_2).
Column 1. $T(v_1) = T(1,1) = (1,2)$. Find a, b with $(1,2) = a(1,1) + b(1,-1) = (a+b,\ a-b)$:

$$\begin{cases} a + b = 1 \\ a - b = 2 \end{cases} \quad \Rightarrow \quad 2a = 3,\ a = \tfrac{3}{2},\ b = 1 - a = -\tfrac{1}{2}.$$

So $T(v_1) = \frac{3}{2}v_1 - \frac{1}{2}v_2$ and $[T(v_1)]_{B'} = (\frac{3}{2}, -\frac{1}{2})^\top$.
Column 2. $T(v_2) = T(1,-1) = (1,-2)$. Find a, b with $(1,-2) = (a+b,\ a-b)$:

$$\begin{cases} a + b = 1 \\ a - b = -2 \end{cases} \quad \Rightarrow \quad 2a = -1,\ a = -\tfrac{1}{2},\ b = 1 - a = \tfrac{3}{2}.$$

So $T(v_2) = -\frac{1}{2}v_1 + \frac{3}{2}v_2$ and $[T(v_2)]_{B'} = (-\frac{1}{2}, \frac{3}{2})^\top$.

$$\boxed{[T]_{B'} = \begin{pmatrix} \frac{3}{2} & -\frac{1}{2} \\ -\frac{1}{2} & \frac{3}{2} \end{pmatrix}}$$

SOLUTION // Ex. 2.2 – method 2 (formula $P^{-1}AP$)

$P = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$ (new basis vectors as columns; it satisfies $[x]_B = P[x]_{B'}$), $A = [T]_B = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}$, and $P^{-1} = \frac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$.

1. $AP = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 2 & -2 \end{pmatrix}$.
2. $P^{-1}(AP) = \frac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}\begin{pmatrix} 1 & 1 \\ 2 & -2 \end{pmatrix} = \frac{1}{2}\begin{pmatrix} 3 & -1 \\ -1 & 3 \end{pmatrix} = \begin{pmatrix} \frac{3}{2} & -\frac{1}{2} \\ -\frac{1}{2} & \frac{3}{2} \end{pmatrix}$. ✓

Same answer. **NB:** the matrix of a linear map depends *in general* on the basis: here a diagonal matrix became non-diagonal. But not always: the identity map has matrix I_n in every basis.

Exercise 3

Let $A \in \mathbb{R}^{m\times n}$, $B \in \mathbb{R}^{n\times p}$. Verify directly that $(AB)_{ij} = \sum_{k=1}^{n} a_{ik}b_{kj}$.

SOLUTION // Ex. 3

Write $A = (a_{ik})_{1\le i\le m,\, 1\le k\le n}$ and $B = (b_{kj})_{1\le k\le n,\, 1\le j\le p}$. The j-th column of B is $B_{\cdot j} = (b_{1j}, b_{2j}, \dots, b_{nj})^\top$.
The product AB is the matrix of "apply B, then A", so its j-th column is $(AB)e_j = A(Be_j) = A\,B_{\cdot j}$: the matrix A applied to the j-th column of B.
The i-th component of the matrix–vector product $A\,B_{\cdot j}$ is the i-th row of A times the column $B_{\cdot j}$:

$$(AB)_{ij} = \begin{pmatrix} a_{i1} & a_{i2} & \cdots & a_{in} \end{pmatrix}\begin{pmatrix} b_{1j} \\ b_{2j} \\ \vdots \\ b_{nj} \end{pmatrix} = a_{i1}b_{1j} + a_{i2}b_{2j} + \cdots + a_{in}b_{nj}.$$

Hence $\boxed{(AB)_{ij} = \sum_{k=1}^{n} a_{ik}b_{kj}}$ for all $1 \le i \le m$, $1 \le j \le p$.

Exercise 4

Compute the rank and kernel of $A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}$, then $\operatorname{rank}(A)$ and $\dim\ker(A)$.

SOLUTION // Ex. 4

A defines the map $\mathbb{R}^3 \to \mathbb{R}^2$, $A(x_1, x_2, x_3)^\top = (x_1, x_2)^\top$ (it forgets the third coordinate).
Rank. *Method 1 (columns):* the three columns $(1,0), (0,1), (0,0)$ are dependent, but the first two are independent, so the column space has dimension 2. *Method 2 (pivots):* A is already in (reduced) row-echelon form with two pivots (columns 1 and 2). Either way $\boxed{\operatorname{rank}(A) = 2}$.

Kernel. Rank–nullity already says $\dim\ker(A) = 3 - 2 = 1$. Precisely: $Ax = 0$ gives $x_1 = 0$, $x_2 = 0$, and x_3 is free. So

$$\ker(A) = \left\{ \begin{pmatrix} 0 \\ 0 \\ t \end{pmatrix} : t \in \mathbb{R} \right\} = \operatorname{span}\left\{ \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \right\}, \qquad \boxed{\dim\ker(A) = 1}.$$

Check: $\operatorname{rank}(A) + \dim\ker(A) = 2 + 1 = 3 =$ number of columns. ✓ The map is surjective (rank 2 = dim of $\mathbb{R}^2$) but not injective.

Exercise 5

Are the following matrices invertible? If so, compute their inverse.

$$A = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad C = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}, \quad D = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 1 & 1 & 1 \end{pmatrix}.$$

SOLUTION // Ex. 5 – A

A is diagonal with non-zero diagonal entries, so it is invertible, and its inverse is obtained by taking the reciprocal of each diagonal entry:

$$\boxed{A^{-1} = \begin{pmatrix} 1 & 0 \\ 0 & \frac{1}{2} \end{pmatrix}} \qquad \text{Check: } \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 0 & \frac{1}{2} \end{pmatrix} = I_2. \checkmark$$

SOLUTION // Ex. 5 – B

$B\begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$ with $(0,1) \neq 0$, so $\ker(B) \neq \{0\}$: $\boxed{B \text{ is not invertible}}$. (Equivalently: zero row, $\operatorname{rank} B = 1 < 2$, a diagonal entry is 0, and $\det B = 1 \cdot 0 - 0 \cdot 0 = 0$.)

SOLUTION // Ex. 5 – C (Gauss–Jordan)

$$\left(\begin{array}{ccc|ccc} 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array}\right)$$

C is already upper triangular with pivots $1, 1, 1$: three pivots, so C is invertible. We only need to clear *above* the pivots.

Step 1. Clear the entry above the pivot of column 3: $R_2 \leftarrow R_2 - R_3$.

$$\left(\begin{array}{ccc|ccc} 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & -1 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array}\right)$$

Step 2. Clear the entry above the pivot of column 2: $R_1 \leftarrow R_1 - R_2$.

$$\left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & -1 & 1 \\ 0 & 1 & 0 & 0 & 1 & -1 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array}\right)$$

The left block is I_3, so the right block is the inverse:

$$\boxed{C^{-1} = \begin{pmatrix} 1 & -1 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix}}$$

Check: $CC^{-1} = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & -1 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & -1+1 & 1-1 \\ 0 & 1 & -1+1 \\ 0 & 0 & 1 \end{pmatrix} = I_3$. ✓

SOLUTION // Ex. 5 – D (Gauss–Jordan)

$$\left(\begin{array}{ccc|ccc} 1 & 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 \\ 1 & 1 & 1 & 0 & 0 & 1 \end{array}\right)$$

Step 1. Clear column 1: $R_2 \leftarrow R_2 - R_1$, $R_3 \leftarrow R_3 - R_1$.

$$\left(\begin{array}{ccc|ccc} 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & -1 & 1 & -1 & 1 & 0 \\ 0 & 0 & 1 & -1 & 0 & 1 \end{array}\right)$$

Step 2. Make the second pivot equal to 1: $R_2 \leftarrow -R_2$.

$$\left(\begin{array}{ccc|ccc} 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & -1 & 1 & -1 & 0 \\ 0 & 0 & 1 & -1 & 0 & 1 \end{array}\right)$$

Three pivots: D is invertible. **Step 3.** Clear above the pivot of column 3: $R_2 \leftarrow R_2 + R_3$.

$$\left(\begin{array}{ccc|ccc} 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & -1 & 1 \\ 0 & 0 & 1 & -1 & 0 & 1 \end{array}\right)$$

Step 4. Clear above the pivot of column 2: $R_1 \leftarrow R_1 - R_2$.

$$\left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 1 & -1 \\ 0 & 1 & 0 & 0 & -1 & 1 \\ 0 & 0 & 1 & -1 & 0 & 1 \end{array}\right) \quad \Longrightarrow \quad \boxed{D^{-1} = \begin{pmatrix} 1 & 1 & -1 \\ 0 & -1 & 1 \\ -1 & 0 & 1 \end{pmatrix}}$$

Check (row by row of DD^{-1}): row 1: $(1+0,\ 1-1,\ -1+1) = (1,0,0)$; row 2: $(1-1,\ 1+0,\ -1+1) = (0,1,0)$; row 3: $(1+0-1,\ 1-1+0,\ -1+1+1) = (0,0,1)$. ✓

Exercise 6

Let $A \in \mathbb{R}^{n\times n}$. Prove the equivalence of: (1) $x \mapsto Ax$ is bijective; (2) $\mathrm{Im}(A) = \mathbb{R}^n$; (3) $\ker(A) = \{0\}$.

SOLUTION // Ex. 6

Let $T : \mathbb{R}^n \to \mathbb{R}^n$, $T(x) = Ax$. We prove the cycle (1) ⇒ (2) ⇒ (3) ⇒ (1), using rank–nullity: $\dim\ker(A) + \mathrm{rank}(A) = n$.

(1) ⇒ (2). If T is bijective, it is in particular surjective: every vector of $\mathbb{R}^n$ is the image of some vector, so $\mathrm{Im}(A) = \mathbb{R}^n$.

(2) $\Rightarrow$ (3). Suppose $\mathrm{Im}(A) = \mathbb{R}^n$. Then $\mathrm{rank}(A) = \dim \mathrm{Im}(A) = n$. By rank–nullity, $\dim \ker(A) = n - \mathrm{rank}(A) = 0$, hence $\ker(A) = \{0\}$.
(3) $\Rightarrow$ (1). Suppose $\ker(A) = \{0\}$. Then T is injective (Lecture 1). Also $\dim \ker(A) = 0$, so by rank–nullity $\mathrm{rank}(A) = n$, i.e. $\dim \mathrm{Im}(A) = n$. Since $\mathrm{Im}(A)$ is an n-dimensional subspace of $\mathbb{R}^n$, $\mathrm{Im}(A) = \mathbb{R}^n$: T is surjective. Injective and surjective: T is bijective.
Therefore $\boxed{T \text{ bijective} \iff \mathrm{Im}(A) = \mathbb{R}^n \iff \ker(A) = \{0\}}$.
Proof shape to remember: for a chain of equivalences, prove a cycle of implications. Three arrows instead of six.

Exercise 7 (*)

Find some examples of 3×3 *orthogonal matrices.*

SOLUTION // Ex. 7

$Q \in \mathbb{R}^{3\times 3}$ is orthogonal if $Q^\top Q = I_3$; equivalently its columns form an orthonormal basis of $\mathbb{R}^3$. So: pick three columns of length 1 that are pairwise orthogonal.

$$\underbrace{Q_1 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}}_{\text{identity}} \qquad \underbrace{Q_2 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}}_{\text{reflection}} \qquad \underbrace{Q_3 = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}}_{\text{permutation}}$$

$$\underbrace{Q_4 = \begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}}_{\text{rotation of angle } \theta \text{ around the 3rd axis}}$$

Check for Q_4: columns $c_1 = (\cos\theta, \sin\theta, 0)$, $c_2 = (-\sin\theta, \cos\theta, 0)$, $c_3 = (0, 0, 1)$. Lengths: $\cos^2\theta + \sin^2\theta = 1$ (twice) and 1. Dot products: $c_1^\top c_2 = -\cos\theta\sin\theta + \sin\theta\cos\theta = 0$, $c_1^\top c_3 = c_2^\top c_3 = 0$. So $Q_4^\top Q_4 = I_3$. ✓
A less obvious one: $Q_5 = \frac{1}{3}\begin{pmatrix} 1 & 2 & 2 \\ 2 & 1 & -2 \\ 2 & -2 & 1 \end{pmatrix}$. Each column has squared length $\frac{1+4+4}{9} = 1$, and e.g. $c_1^\top c_2 = \frac{2+2-4}{9} = 0$, $c_1^\top c_3 = \frac{2-4+2}{9} = 0$, $c_2^\top c_3 = \frac{4-2-2}{9} = 0$. ✓
Remark: orthogonal matrices have determinant 1 or -1: rotations have determinant 1, reflections may have determinant -1.

9.2 Part 1(b): True or False

SOLUTION // T/F 1 – "$A \in \mathbb{R}^{m\times n}$ is invertible if and only if $m = n$"

FALSE. Being square is *necessary* for an ordinary two-sided inverse, but not *sufficient*. Counterexample: $A = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$ is square but not invertible ($Ae_2 = 0$). For a square matrix: A invertible $\iff \mathrm{rank}(A) = n \iff \ker(A) = \{0\}$.

SOLUTION // T/F 2 – "If A is diagonal, then A^2 is diagonal"

TRUE. If $A = \mathrm{diag}(\lambda_1, \ldots, \lambda_n)$ then $A^2 = \mathrm{diag}(\lambda_1^2, \ldots, \lambda_n^2)$ (diagonal matrices multiply entry by entry), which is diagonal.

SOLUTION // T/F 3 – "If A^2 is diagonal, then A is diagonal"

FALSE. Counterexample: $A = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ is not diagonal, but $A^2 = \begin{pmatrix} 0\cdot 0+1\cdot 1 & 0\cdot 1+1\cdot 0 \\ 1\cdot 0+0\cdot 1 & 1\cdot 1+0\cdot 0 \end{pmatrix} = I_2$ is diagonal. (Swapping twice does nothing.)

SOLUTION // T/F 4 – "If a linear map $T : \mathbb{R}^n \to \mathbb{R}^n$ is injective, then it is surjective"

TRUE. T injective gives $\ker(T) = \{0\}$, so $\dim\ker T = 0$. By rank–nullity, $\operatorname{rank}(T) + \dim\ker(T) = n$, so $\operatorname{rank}(T) = n$, i.e. $\dim\operatorname{Im}(T) = n$. Since $\operatorname{Im}(T) \subseteq \mathbb{R}^n$, this forces $\operatorname{Im}(T) = \mathbb{R}^n$: T is surjective.
This relies on the domain and codomain having the same finite dimension. For $T : \mathbb{R}^2 \to \mathbb{R}^3$ it would be false.

SOLUTION // T/F 5 – "If $T : \mathbb{R}^n \to \mathbb{R}^n$ is surjective, then its matrix in the standard basis is invertible"

TRUE. T surjective means $\operatorname{Im}(T) = \mathbb{R}^n$, so $\operatorname{rank}(T) = n$. The matrix $A \in \mathbb{R}^{n\times n}$ representing T then has full rank, $\operatorname{rank}(A) = n$, and a square matrix of full rank is invertible (big theorem, item 3).

9.3 Part 2, Exercise 1: matrix representation and change of basis

Let $T : \mathbb{R}^3 \to \mathbb{R}^4$, $T(x_1, x_2, x_3) = (x_1 + 2x_2,\ x_2 - x_3,\ 3x_1 + x_3,\ 2x_1 - x_2 + x_3)$. B is the standard basis of $\mathbb{R}^3$, C the standard basis of $\mathbb{R}^4$.

SOLUTION // 1. Matrix of T in the standard bases

The columns are the images of the standard basis vectors:

$$T(e_1) = T(1,0,0) = \begin{pmatrix} 1 \\ 0 \\ 3 \\ 2 \end{pmatrix}, \qquad T(e_2) = T(0,1,0) = \begin{pmatrix} 2 \\ 1 \\ 0 \\ -1 \end{pmatrix}, \qquad T(e_3) = T(0,0,1) = \begin{pmatrix} 0 \\ -1 \\ 1 \\ 1 \end{pmatrix}.$$

$$\boxed{[T]_{C\leftarrow B} = A = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & -1 \\ 3 & 0 & 1 \\ 2 & -1 & 1 \end{pmatrix}}$$

Size 4×3: 4 rows (output in $\mathbb{R}^4$), 3 columns (input in $\mathbb{R}^3$). ✓ Indeed $[T(x)]_C = A[x]_B$: row i of A holds the coefficients of x_1, x_2, x_3 in the i-th output, e.g. row 4 is $(2, -1, 1)$ for $2x_1 - x_2 + x_3$.

SOLUTION // 2. Uniqueness of the matrix representation

Suppose A and A' both represent T in the same standard bases. Then for every standard basis vector e_j: $Ae_j = T(e_j) = A'e_j$. But Ae_j is the j-th column of A and $A'e_j$ the j-th column of A'. So every column of A equals the corresponding column of A': $A = A'$. The matrix of T in fixed bases is unique.

SOLUTION // 3. Rank and kernel

Gaussian elimination on A.
Step 1. Clear column 1: $R_3 \leftarrow R_3 - 3R_1$, $R_4 \leftarrow R_4 - 2R_1$.

$$\begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & -1 \\ 3 & 0 & 1 \\ 2 & -1 & 1 \end{pmatrix} \longrightarrow \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & -1 \\ 0 & -6 & 1 \\ 0 & -5 & 1 \end{pmatrix}$$

Step 2. Clear column 2 (above and below the pivot): $R_1 \leftarrow R_1 - 2R_2$, $R_3 \leftarrow R_3 + 6R_2$, $R_4 \leftarrow R_4 + 5R_2$.

$$\begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & -1 \\ 0 & 0 & -5 \\ 0 & 0 & -4 \end{pmatrix}$$

Step 3. Clear below the pivot of column 3: $R_4 \leftarrow 5R_4 - 4R_3$ (gives $-20 + 20 = 0$).

$$\begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & -1 \\ 0 & 0 & -5 \\ 0 & 0 & 0 \end{pmatrix}$$

Three pivots: $\boxed{\operatorname{rank}(A) = 3}$.
Kernel. There is a pivot in every column, so there's no free variable and $Ax = 0$ has only the trivial solution: $\boxed{\ker(A) = \{(0,0,0)^\top\}}$, i.e. $\ker(T) = \{0\}$. **Check:** $\operatorname{rank} + \dim\ker = 3 + 0 = 3$ columns. ✓

SOLUTION // 4. Injective, surjective, bijective?

- $\ker(T) = \{0\}$, so T is **injective**.
- $\operatorname{rank}(T) = 3$ but the codomain has dimension $\dim(\mathbb{R}^4) = 4$, so $\operatorname{range}(T) \neq \mathbb{R}^4$: T is **not surjective**. (No surprise: a tall 4×3 matrix has rank at most 3.)

$\boxed{T \text{ is injective, not surjective, hence not bijective.}}$

SOLUTION // 5. Matrix of T in the bases B' and C

$B' = (v_1, v_2, v_3)$ with $v_1 = (1,1,0)$, $v_2 = (0,1,1)$, $v_3 = (1,0,1)$; the codomain keeps the standard basis C. The columns of $[T]_{C \leftarrow B'}$ are $T(v_1), T(v_2), T(v_3)$ (already in standard coordinates, nothing to convert):

$$\begin{aligned} T(v_1) &= T(1,1,0) = (1+2,\ 1-0,\ 3+0,\ 2-1+0) = (3,1,3,1), \\ T(v_2) &= T(0,1,1) = (0+2,\ 1-1,\ 0+1,\ 0-1+1) = (2,0,1,0), \\ T(v_3) &= T(1,0,1) = (1+0,\ 0-1,\ 3+1,\ 2-0+1) = (1,-1,4,3). \end{aligned}$$

$$\boxed{[T]_{C \leftarrow B'} = \begin{pmatrix} 3 & 2 & 1 \\ 1 & 0 & -1 \\ 3 & 1 & 4 \\ 1 & 0 & 3 \end{pmatrix}}$$

SOLUTION // 6. Change-of-basis matrix and verification of the formula

Careful with notation. We use $P_{B\leftarrow B'}$ for the matrix whose columns are the vectors of B' written in the standard basis B:

$$P_{B\leftarrow B'} = \begin{pmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix}, \qquad [x]_B = P_{B\leftarrow B'}[x]_{B'}.$$

The other direction, $P_{B'\leftarrow B}$. Let $x = (x_1, x_2, x_3)$ and $[x]_{B'} = (a, b, c)^\top$, i.e. $x = av_1 + bv_2 + cv_3$:

$$\begin{cases} a + c = x_1 \\ a + b = x_2 \\ b + c = x_3 \end{cases}$$

1. (eq. 1) + (eq. 2) − (eq. 3): $2a = x_1 + x_2 - x_3$, so $a = \dfrac{x_1 + x_2 - x_3}{2}$.
2. (eq. 2) + (eq. 3) − (eq. 1): $2b = -x_1 + x_2 + x_3$, so $b = \dfrac{-x_1 + x_2 + x_3}{2}$.
3. (eq. 1) + (eq. 3) − (eq. 2): $2c = x_1 - x_2 + x_3$, so $c = \dfrac{x_1 - x_2 + x_3}{2}$.

Reading the coefficients row by row:

$$\boxed{P_{B'\leftarrow B} = \frac{1}{2}\begin{pmatrix} 1 & 1 & -1 \\ -1 & 1 & 1 \\ 1 & -1 & 1 \end{pmatrix}}, \qquad [x]_{B'} = P_{B'\leftarrow B}[x]_B.$$

Always check: $P_{B'\leftarrow B}P_{B\leftarrow B'} = \frac{1}{2}\begin{pmatrix} 1 & 1 & -1 \\ -1 & 1 & 1 \\ 1 & -1 & 1 \end{pmatrix}\begin{pmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix} = \frac{1}{2}\begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix} = I_3$. ✓ So $P_{B'\leftarrow B} = (P_{B\leftarrow B'})^{-1}$.

Verifying the change-of-basis formula. The basis of the codomain has not changed, so

$$[T]_{C\leftarrow B'} = [T]_{C\leftarrow B}\, P_{B\leftarrow B'}$$

(feed B'-coordinates, convert them to B-coordinates, then apply A). Indeed:

$$\begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & -1 \\ 3 & 0 & 1 \\ 2 & -1 & 1 \end{pmatrix}\begin{pmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix} = \begin{pmatrix} 1+2 & 2 & 1 \\ 1 & 1-1 & -1 \\ 3 & 1 & 3+1 \\ 2-1 & -1+1 & 2+1 \end{pmatrix} = \begin{pmatrix} 3 & 2 & 1 \\ 1 & 0 & -1 \\ 3 & 1 & 4 \\ 1 & 0 & 3 \end{pmatrix},$$

exactly the matrix of question 5. ✓

SOLUTION // 7. Are the two matrix representations similar?

Trick question. No notion of similarity applies here. Two matrices $A, \widetilde{A}$ are similar if $\widetilde{A} = P^{-1}AP$ for an invertible P: this is a notion for **square** matrices, which arises when an *endomorphism* $T : V \to V$ is written in two bases of the *same* space. Here $T : \mathbb{R}^3 \to \mathbb{R}^4$ and the matrices are 4×3.

The two representations *are* related by a change of basis, $[T]_{C\leftarrow B'} = I_4\,[T]_{C\leftarrow B}\,P_{B\leftarrow B'}$, which makes them **equivalent** matrices ($B = QAP$ with $Q = I_4$ and $P = P_{B\leftarrow B'}$ invertible), but not similar.

9.4 Part 2, Exercise 2: rank of a matrix

Let $A = \begin{pmatrix} 0 & 1 & 2 \\ -1 & 0 & 3 \\ 1 & 1 & 0 \end{pmatrix}$. *(1) Rank by row reduction. (2) Basis of the column space. (3) Basis of the null space.*

SOLUTION // Row reduction first, then answer everything

The top-left entry is 0, so we need a swap to get a pivot.

Step 1. $R_1 \leftrightarrow R_2$: $\begin{pmatrix} -1 & 0 & 3 \\ 0 & 1 & 2 \\ 1 & 1 & 0 \end{pmatrix}$

Step 2. Clear column 1: $R_3 \leftarrow R_3 + R_1$: $\begin{pmatrix} -1 & 0 & 3 \\ 0 & 1 & 2 \\ 0 & 1 & 3 \end{pmatrix}$

Step 3. Clear column 2: $R_3 \leftarrow R_3 - R_2$: $\begin{pmatrix} -1 & 0 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{pmatrix}$ (row-echelon form, three pivots)

Step 4. Continue to the reduced form. $R_1 \leftarrow -R_1$: $\begin{pmatrix} 1 & 0 & -3 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{pmatrix}$

Step 5. Clear above the pivot of column 3: $R_1 \leftarrow R_1 + 3R_3$, $R_2 \leftarrow R_2 - 2R_3$:

$$\operatorname{rref}(A) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = I_3.$$

SOLUTION // 1. Rank

Three pivot positions: $\boxed{\operatorname{rank}(A) = 3}$.

SOLUTION // 2. Basis of the column space

Every column is a pivot column, so the three *original* columns form a basis of the column space:

$$\boxed{\left\{ \begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 2 \\ 3 \\ 0 \end{pmatrix} \right\} \text{ is a basis of } \operatorname{Col}(A).}$$

Since the rank is 3, $\operatorname{Col}(A) = \mathbb{R}^3$ (so the standard basis would be an equally valid answer).

SOLUTION // 3. Basis of the null space

Method 1 (rank–nullity): $\dim\ker(A) = 3 - \operatorname{rank}(A) = 0$. *Method 2 (solve* $Ax = 0$*):* the reduced form is I_3, so $x_1 = x_2 = x_3 = 0$. Either way

$$\boxed{\ker(A) = \{0\}}, \qquad \boxed{\dim\ker(A) = 0}.$$

Important: the zero vector space has the **empty family** as a basis: a basis of $\ker(A)$ is $\varnothing$. (Not $\{0\}$: a family containing the zero vector is never independent.)

SOLUTION // Bonus (one more for practice): are the columns of A linearly independent?

Yes. $\ker(A) = \{0\}$ means the only combination of the columns giving 0 is the trivial one. Equivalently: 3 columns and rank 3. By the big theorem, A is also invertible, its rows are independent, and $Ax = b$ has a unique solution for every $b \in \mathbb{R}^3$.

10 Final boss: the cheat sheet

10.1 "If they ask you this, do that"

If asked...	**Do this**
Compute AB	Check sizes $(m \times n)(n \times p)$. Entry (i, j) = row i of A times column j of B. Result is $m \times p$.
Matrix of T in standard bases	Columns $= T(e_1), \ldots, T(e_n)$. Or: row i = coefficients of the i-th output formula.
Matrix of T in bases $\mathcal{B}, \mathcal{C}$	Compute $T(v_j)$ for each $v_j \in \mathcal{B}$, write it in the basis $\mathcal{C}$, put the coordinates in column j.
Matrix of T in a new basis $\mathcal{B}'$ (endomorphism)	P = new basis vectors as columns; answer $= P^{-1}AP$. Or do it by the definition.
Coordinates of x in a new basis	Solve $Pc = x$; then $c = [x]_{\mathcal{B}'}$.
Rank of A	Row-reduce, count pivots.
Kernel of A	Solve $Ax = 0$ by elimination; one basis vector per free variable.
Basis of the column space	Pivot columns of the **original** A.
Basis of the row space	Non-zero rows of the echelon form.
Is $x \mapsto Ax$ injective?	$\ker A = \{0\}$? i.e. a pivot in every column ($\operatorname{rank} A = n$). Never if A is wide.
Is $x \mapsto Ax$ surjective?	$\operatorname{rank} A = m$? i.e. a pivot in every row. Never if A is tall.
Is A invertible?	Square AND full rank. Use any item of the big theorem. Not invertible: exhibit $x \neq 0$ with $Ax = 0$.
Compute A^{-1}	Diagonal: reciprocals. 2×2: formula. Orthogonal: transpose. Otherwise Gauss–Jordan on $[A \mid I]$. Then check $AA^{-1} = I$.
Solve $Ax = b$	Row-reduce $[A \mid b]$. Row $(0 \cdots 0 \mid c \neq 0)$: no solution. Else back-substitute; free variables give parameters. Answer $= x_p + \ker A$.
How many solutions?	Compare $\operatorname{rank} A$, $\operatorname{rank}[A \mid b]$ and n.
Are A, B similar?	Must be square, same size, with $B = P^{-1}AP$. Non-square: the question doesn't apply (maybe *equivalent*).
Show two things are equivalent	Prove a cycle of implications $(1) \Rightarrow (2) \Rightarrow (3) \Rightarrow (1)$.
True/False	True: prove it in general. False: one explicit small counterexample (2×2 matrices are your friends).

10.2 Formula sheet

Product	$(AB)_{ij} = \sum_{k=1}^{n} a_{ik}b_{kj}, \quad Ax = x_1a_1 + \cdots + x_na_n, \quad AB = [\,Ab_1 \mid \cdots \mid Ab_p\,]$
Transpose	$(A^\top)^\top = A, \quad (A+B)^\top = A^\top + B^\top, \quad (\alpha A)^\top = \alpha A^\top, \quad (AB)^\top = B^\top A^\top$
Inverse	$(A^{-1})^{-1} = A, \quad (AB)^{-1} = B^{-1}A^{-1}, \quad (A^\top)^{-1} = (A^{-1})^\top, \quad (\alpha A)^{-1} = \frac{1}{\alpha}A^{-1}$
2×2 inverse	$\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad-bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$ if $ad - bc \neq 0$
Rank	$\operatorname{rank} A = \operatorname{rank} A^\top \leq \min(m, n), \quad \operatorname{rank}(AB) \leq \min(\operatorname{rank} A, \operatorname{rank} B)$
Rank–nullity	$n = \operatorname{rank}(A) + \dim \ker(A) \quad (n = \text{number of columns})$
Matrix of a map	$[L(x)]_{\mathcal{C}} = [L]_{\mathcal{C}\leftarrow\mathcal{B}}[x]_{\mathcal{B}}, \quad [M \circ L] = [M][L]$
Change of basis	$[x]_{\mathcal{B}} = P[x]_{\mathcal{B}'}, \quad [L]_{\mathcal{B}'} = P^{-1}[L]_{\mathcal{B}}P, \quad P = \text{new basis in old coordinates}$
Diagonal	$\operatorname{diag}(d_i)\operatorname{diag}(d_i') = \operatorname{diag}(d_id_i'), \quad \operatorname{diag}(d_i)^{-1} = \operatorname{diag}(1/d_i)$
Orthogonal	$U^\top U = UU^\top = I, \quad U^{-1} = U^\top$
Least squares	$\hat{\beta} = (X^\top X)^{-1}X^\top y \quad (\text{when } X^\top X \text{ is invertible})$

10.3 Hall of Ls (classic mistakes)

COMMON L (don't do this) // Do not

1. Assume $AB = BA$. It's false in general, and sometimes only one of the two products even exists.
2. Confuse rows and columns: $m \times n = m$ rows, n columns; a_{ij} = row i, column j; $A \in \mathbb{R}^{m\times n}$ maps $\mathbb{R}^n \to \mathbb{R}^m$.
3. "Simplify" matrices like numbers: $AB = 0$ doesn't give $A = 0$ or $B = 0$; $AB = AC$ doesn't give $B = C$.
4. Forget that transposes and inverses **reverse the order**: $(AB)^\top = B^\top A^\top$, $(AB)^{-1} = B^{-1}A^{-1}$.
5. Invert entry by entry, or say a square matrix is automatically invertible.
6. Use rank–nullity with the number of **rows**. It's the number of **columns**.
7. Take the pivot columns from the *reduced* matrix as a basis of the column space. Take them from the original A.
8. Mix up the direction of P: columns = *new* basis in *old* coordinates, and $[x]_{\text{old}} = P[x]_{\text{new}}$.
9. Talk about "similar" matrices that aren't square.
10. Multiply a row by 0 during elimination, or forget to apply the operation to the right-hand side.
11. Say a basis of $\{0\}$ is $\{0\}$. It's the empty family.
12. Skip the final check. $AA^{-1} = I$ and "plug the solution back in" take 30 seconds and catch most errors.

10.4 The whole lecture in 10 lines

TL;DR // Screenshot this

1. **Matrix** = rectangle of numbers, m rows $\times$ n columns. Add and scale entry by entry.
2. **Product**: row times column; sizes $(m \times n)(n \times p) = (m \times p)$; associative, **not commutative**.
3. Ax **= combination of the columns of** A**.** So range A = span of the columns.
4. **Matrix = linear map.** Columns = images of the basis vectors. Composition = product.
5. **Rank** = dim of column space = dim of row space = number of pivots. **Kernel** = solutions of $Ax = 0$. **Rank + nullity = number of columns.**
6. **Special matrices**: diagonal (rescale), identity (nothing), symmetric ($A = A^\top$), orthogonal ($U^{-1} = U^\top$), sparse, triangular…
7. **Invertible** $\iff$ $\ker = \{0\}$ $\iff$ full rank $\iff$ independent columns $\iff$ $Ax = b$ always has a unique solution $\iff$ reduces to I_n.
8. **Change of basis**: $[x]_{\text{old}} = P[x]_{\text{new}}$ and $[L]_{\text{new}} = P^{-1}[L]_{\text{old}}P$. Similar matrices = same map, different basis.
9. **Linear systems** $Ax = b$: 0, 1 or ∞ solutions; general solution = particular + kernel. **Gaussian elimination** solves everything: systems, rank, kernel, inverse.
10. **ML**: layers are $Wx + b$, batches are matrix products, kernel = lost information, LoRA = low-rank update, covariance = symmetric, convolution = structured sparse matrix, regression = (least-squares) linear system.

Want more practice?

Good resources for more practice: the free textbook (and lecture notes) *Mathematics for Machine Learning* (Section 2.1 onwards, lots of linear systems to solve); the Matrix Cookbook for formulas; and d2l.ai for the same ideas with executable code.

Next episode: determinants (Lecture 3). Stay tuned, bestie.